

Quantum vs. Classical Correlations in Dijet Hadron Multiplicities: A Finite-Statistics Methodology Note

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July 22, 2026

Abstract

We develop and validate a complete methodological pipeline for measuring mutual information (MI), conditional entropy $S(B|A)$, and a normalized witness $\text{NMI} = I(A:B)/\min(S_A, S_B)$ between back-to-back jet hadron multiplicities, testing whether an observed dijet correlation is classically explicable or could signal genuine quantum entanglement. The naive plug-in entropy estimator is biased upward at finite sample size, in exactly the direction that could mimic a quantum excess; using a Monte Carlo toy model with known ground truth, we validate two standard corrections (Miller–Madow, shuffle-subtraction) and a Bayesian estimator (NSB), and derive a quantitative detection-sensitivity curve from the best-behaved of these. The full pipeline, cross-validated between independent Python and ROOT/C++ implementations, is applied to realistic PYTHIA 8 dijet Monte Carlo. Since PYTHIA output carries no independently known true MI, we develop a closure test that calibrates a synthetic model to the data’s own statistics; applied to a full validation sample this finds NSB’s bootstrap uncertainty under-covers by roughly $1.5\text{--}1.7\times$, improving substantially when the same test is run separately within p_T bins. We further identify and correct a distinct systematic: any finite p_T window induces real but kinematically trivial MI, since both jets’ multiplicities and their p_T ’s are correlated through momentum balance. A mixed-event subtraction – pairing each jet with a p_T -matched partner from a different event – isolates the beyond-kinematics MI; closure tests, both on synthetic data and at full statistics on the real sample’s own fitted kinematics, show the naive estimator’s finite-sample bias nearly cancels between the same-event and mixed-event terms, so the raw difference is close to unbiased even before rigorous correction. Applied to the real validation sample, the subtraction finds a nonzero beyond-kinematics component of order 0.003 nats – consistent across all four estimator methods and plausibly attributable to shared underlying-event activity rather than anything exotic. All uncertainties throughout are computed by resampling composite quantities directly rather than propagating component errors separately, a distinction we show matters in practice. The result is a validated, closure-tested toolkit – estimator, sensitivity curve, calibration factors, and kinematic-background subtraction – ready to apply to real STAR/ALICE dijet data.

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1 Physics motivation

Take a back-to-back dijet event (e.g. from e^+e^- , DIS, or pp collisions at RHIC/STAR). Treat jet A and jet B as the two subsystems of a bipartite system whose “state” is accessed only through ob-

served hadron multiplicities, via **local parton–hadron duality (LPHD)**: the hadron multiplicity distribution stands in for the underlying parton Fock-state distribution.

The Schmidt-decomposition ansatz (Kharzeev–Levin; Peschanski–Seki): if each Fock/parton-number sector n carries probability $P(n)$, and the reduced density matrix of a jet is diagonal in n , then the entanglement entropy of that jet reduces to the **Shannon entropy of its hadron multiplicity distribution**,

$$S_A = - \sum_n P_A(n) \ln P_A(n), \quad (1)$$

and likewise for S_B . This turns a genuinely quantum quantity (entanglement entropy) into something directly measurable from ordinary multiplicity counting – the whole program rests on this ansatz.

The physics question: by additionally measuring the *joint* multiplicity distribution $P_{AB}(n_A, n_B)$ across both jets in the same event, we can construct the mutual information $I(A:B)$ between the two jets (defined in Sec. 2). A large enough MI, in the right combination with the individual entropies, would signal correlations between the jets that cannot be explained by any classical joint probability distribution – i.e. a genuine quantum (entanglement) signature surviving from the partonic state through hadronization, rather than an ordinary classical correlation (e.g. shared color flow, momentum conservation, or other QCD-driven but fundamentally classical correlations).

Why this is subtle in practice: any such claim is only as good as (a) correctly separating genuine quantum correlation from unavoidable classical correlation, and (b) correctly accounting for the finite size of real data samples, since MI estimated from a finite event sample is **always statistically biased**, and – as we show below – biased in exactly the direction that could be mistaken for a quantum excess if not handled carefully. This note works through both issues end to end.

2 The observable: MI, conditional entropy, and the quantum witness

From the per-jet and joint hadron-multiplicity distributions we can build three entropies and two derived quantities:

$$S_A, \quad S_B, \quad S_{AB} \quad \implies \quad I(A:B) = S_A + S_B - S_{AB}, \quad S(B|A) = S_{AB} - S_A. \quad (2)$$

Mutual information alone cannot distinguish quantum from classical. $I(A:B) > 0$ just means the jets are correlated *somehow* – a purely classical joint distribution (shared color flow, energy–momentum conservation, any ordinary QCD dynamics) also gives nonzero MI. A sharper diagnostic is needed.

The sharper test: conditional entropy negativity. For *any* classical discrete probability distribution, Shannon conditional entropy obeys

$$S(B|A) = S_{AB} - S_A \geq 0 \quad (\text{always, classically}). \quad (3)$$

This is the classical subadditivity of Shannon entropy – not an assumption about any particular physical model. Only the *quantum* (von Neumann) conditional entropy can go negative, and only for entangled states. So $S(B|A) < 0$ is a **hard classical impossibility**, and observing it (at sufficient statistical significance) would be direct evidence that the jets cannot be described by a

classical joint probability distribution – i.e. a genuine quantum signature. Equivalently, in terms of MI alone,

$$I(A:B) > \min(S_A, S_B) \iff S(B|A) < 0 \text{ or } S(A|B) < 0, \quad (4)$$

so MI exceeding the smaller of the two marginal entropies is the same statement in a different form. Both directions ($S(B|A)$ and $S(A|B)$) are worth checking independently, since conditional entropy is not symmetric.

A scale-comparable form: normalized mutual information. Absolute $I(A:B)$ is awkward to compare across different collision energies, kinematic selections, or p_T bins, since S_A and S_B themselves grow with e.g. jet p_T (Sec. 9.2) and so set a different natural scale for I in every bin. The $\min(S_A, S_B)$ appearing above already does the needed normalization:

$$\text{NMI} \equiv \frac{I(A:B)}{\min(S_A, S_B)}. \quad (5)$$

Classically $\text{NMI} \in [0, 1]$ always, by the same subadditivity argument as $S(B|A) \geq 0$; $\text{NMI} > 1$ is the identical classical-bound violation as $S(B|A) < 0$, just expressed as a ratio rather than a difference. Since NMI is dimensionless and bounded, it is directly comparable across p_T bins, run periods, or even experiments in a way raw $I(A:B)$ is not, while remaining exactly as valid a quantum witness as $S(B|A)$ itself. This is a standard normalization in the wider MI literature (one of several conventional choices, e.g. Kvalseth 1987; min-normalization is the tightest of these and the one that coincides exactly with the classical bound already in use here).

What this test cannot do: genuine quantum discord. True (Ollivier–Zurek) quantum discord is defined by optimizing conditional entropy over *quantum measurement bases* – non-commuting projective measurements that don’t reduce to simply reading off a classical value. Hadron-multiplicity data (N_A, N_B) is, by construction, a classical joint distribution (a histogram). For any classical joint distribution, the optimal “measurement” on B is trivially just observing N_B exactly, which makes the classical-correlation piece $J(A:B)$ equal to the *full* $I(A:B)$ exactly – so discord $D = I - J \equiv 0$ identically, for any classical distribution, with no exception. Extracting a genuinely nonzero discord requires actual quantum-state tomography (e.g. the $\Lambda\bar{\Lambda}$ spin-density-matrix channel), not multiplicity counting. **The conditional-entropy-negativity test above is therefore the correct and complete witness available from this kind of data** – the rest of this note is about measuring it reliably.

3 The statistical problem: finite-sample bias

Every quantity above is estimated from a finite event sample, and the naive (“plug-in”) entropy estimator has a **systematic, structural bias** that turns out to point in exactly the dangerous direction for this measurement.

Where the bias comes from. The naive estimator replaces true probabilities p_i with empirical frequencies $\hat{p}_i = n_i/N$,

$$\hat{S} = - \sum_i \hat{p}_i \ln \hat{p}_i. \quad (6)$$

Because $x \ln x$ is convex (equivalently $-x \ln x$ is concave), Jensen’s inequality guarantees

$$\langle \hat{S} \rangle < S_{\text{true}} \quad \text{for any finite } N, \quad (7)$$

i.e. the plug-in estimator always underestimates entropy. This isn’t a numerical artifact; with only N draws one can never fully resolve a distribution’s tails, and that missing resolution always looks like *less* apparent randomness than there truly is.

Leading-order size of the effect (Miller 1955; Madow), for a distribution over K possible outcomes:

$$\langle \hat{S} \rangle \approx S_{\text{true}} - \frac{K-1}{2N}. \quad (8)$$

Fewer events (N small) or finer binning (K large) both make the downward bias worse.

Why this flips sign and biases $I(A:B)$ upward. Combining the individual biases of $\hat{S}_A, \hat{S}_B, \hat{S}_{AB}$ with correct signs,

$$\langle \hat{I} \rangle \approx I_{\text{true}} + \frac{(K_{AB}-1) - (K_A-1) - (K_B-1)}{2N} = I_{\text{true}} + \frac{K_{AB} - K_A - K_B + 1}{2N}. \quad (9)$$

The joint distribution lives in a much larger space than either marginal alone, so $\hat{S}_{AB}$ suffers a *much larger* downward bias than $\hat{S}_A + \hat{S}_B$ does – and since S_{AB} enters with a **minus** sign, its extra negative bias flips into a **positive** bias on $\hat{I}$. In the extreme case $N \ll K_A K_B$ (badly undersampled joint histogram), this produces large *phantom* apparent mutual information even for truly independent variables – sampling noise alone can masquerade as a real correlation, and by the same mechanism can masquerade as evidence for the “quantum” side of the $S(B|A) < 0$ test above if not corrected.

A commonly-quoted shortcut version of Eq. (9) is $(K_A-1)(K_B-1)/(2N)$; this is not a different formula, it is Eq. (9) evaluated under the extra assumption $K_{AB} = K_A K_B$, i.e. a fully and uniformly occupied joint space. That assumption only holds for independent variables; for correlated data the observed K_{AB} is much smaller than $K_A K_B$ (the joint distribution concentrates on a narrow ridge rather than filling the full grid), so this shortcut badly *overestimates* the bias – a pitfall we hit directly in Sec. 5.

Practical consequence. Comparing the same underlying physics across data samples of different size – different STAR running periods, different luminosity, different jet-cone or multiplicity binning choices – will report systematically different central MI (and $S(B|A)$) values purely from this effect, with zero change in the actual QCD physics. Secs. 4–6 demonstrate this concretely with a toy model and work through three ways to correct for it; Sec. 7 turns the corrected, best-behaved estimator into a quantitative statement about how large a genuine quantum signal would need to be, at a given event count, to be distinguishable from this bias.

4 Toy Monte Carlo test bed

To demonstrate the bias concretely and validate corrections against a known ground truth, we use a controlled toy model: two back-to-back jets, A and B , with a shared latent hard-process fluctuation Λ (e.g. fluctuating parton virtuality / initial-state energy scale) drawn from a Gamma distribution. Conditional on Λ , each jet’s hadron multiplicity is Poisson distributed,

$$\Lambda \sim \text{Gamma}(k, \theta), \quad N_A | \Lambda \sim \text{Poisson}(\Lambda), \quad N_B | \Lambda \sim \text{Poisson}(c\Lambda). \quad (10)$$

Marginalizing over Λ gives each jet a **Negative Binomial** multiplicity distribution (the standard KNO-like form used for hadron multiplicities), and the shared Λ induces a genuine – but purely **classical** – correlation between N_A and N_B , i.e. $I_{\text{true}} > 0$ with $S(B|A)_{\text{true}}$ solidly positive, exactly as required classically. This is a convenient sandbox: we know the true joint distribution (to arbitrary precision, from a huge reference sample), so we can measure the *bias* of any estimator directly, and later measure how large a genuine quantum excess would need to be to be distinguishable from this classical baseline’s own statistical noise. Parameters used throughout: $k = 3.0$, $\theta = 8.0$, $c = 0.9$ (giving $\text{corr}(N_A, N_B) \approx 0.88$).

4.1 Demonstration: two samples, same true distribution, different size

We generate a “small” event sample ($N = 500$; e.g. a single limited run / low-luminosity STAR-like dataset), a “large” event sample ($N = 200,000$; high-statistics dataset), and a very large reference sample ($N = 5 \times 10^6$) to pin down I_{true} – all drawn from the *exact same underlying physics*.

Table 1: Naive $\hat{I}$ for three samples of the same true physics.

Sample	N	K_A	K_B	K_{AB}	$\hat{I}_{\text{naive}}$ (nats)
small	500	67	60	390	1.7424
large	200,000	129	113	3949	0.7463
reference	5,000,000	159	147	7022	0.7329 ($\approx I_{\text{true}}$)

We see $\hat{I}_{\text{small}} > \hat{I}_{\text{large}} \approx \hat{I}_{\text{ref}} \equiv I_{\text{true}}$ – **the same physics, analyzed with fewer events, reports a systematically larger MI**. This is the bias predicted in Sec. 3, not a change in physics.

The bias comes entirely from how well-populated the joint histogram is. Figure 1 shows (top) the marginal hadron-multiplicity distributions $P_A(n)$, overlaid for the three samples – these look reasonably similar even at $N = 500$ because there are only $O(10^2)$ bins to fill – and (bottom) the 2D joint histograms $P_{AB}(n_A, n_B)$ on the same color scale, where the difference is dramatic: at $N = 500$ the joint plane is almost empty, while at $N = 5 \times 10^6$ it is a smooth, fully-resolved correlated ridge. A sparse, “salt-and-pepper” joint histogram is exactly what drives $\hat{S}_{AB}$ artificially low, and hence $\hat{I}$ artificially high.

4.2 Bias vs. sample size

Repeating the naive estimate over a range of N (with several repetitions per N to see the scatter) and plotting $\hat{I}_{\text{naive}}$ against $1/N$ (Fig. 2) shows the systematic downward drift (bias shrinking) as N grows, converging toward the flat reference line, and the expected near-linear behavior in $1/N$: the intercept of that line is a cheap, model-independent estimate of the bias-corrected/true MI. The extrapolated intercept, 0.9427 nats, lands reasonably close to the reference value of 0.7329 nats.

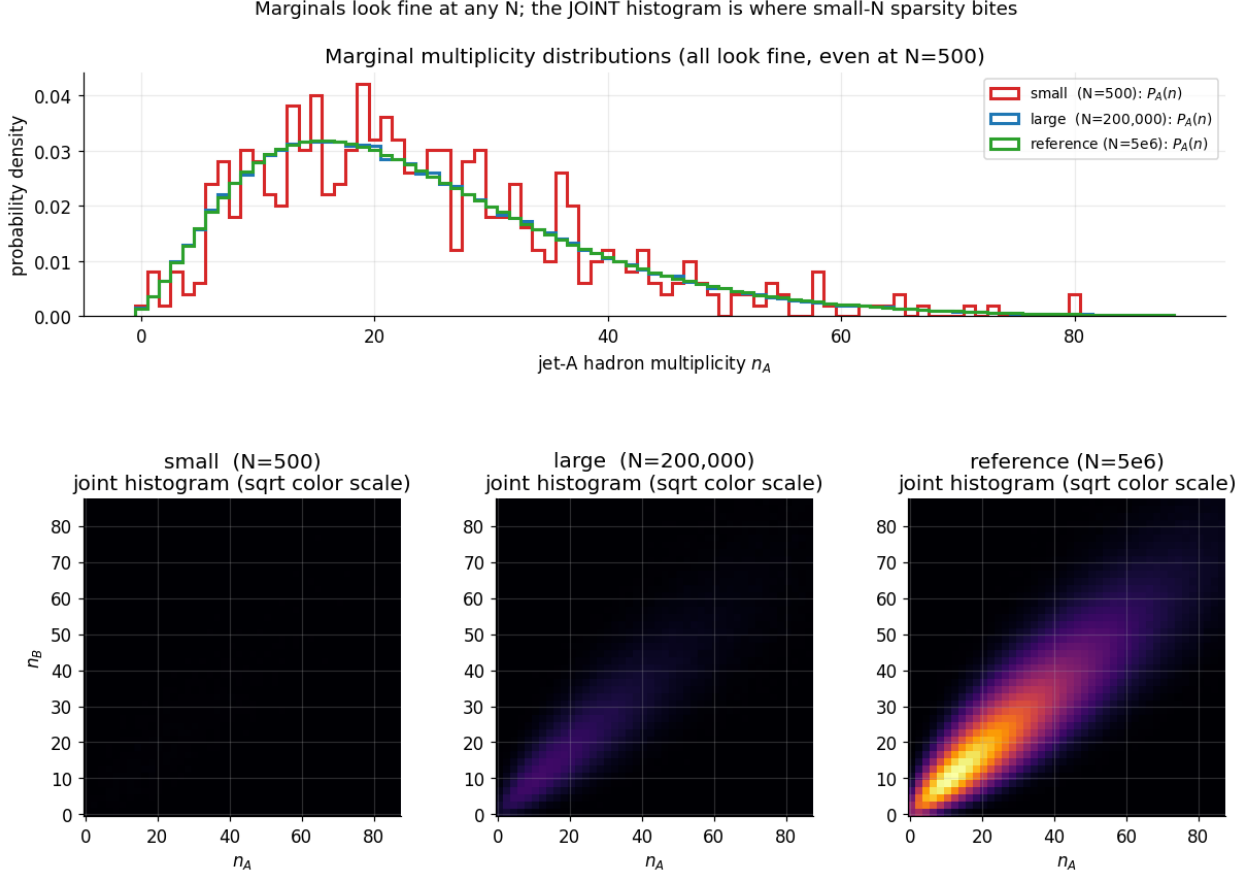


Figure 1: Marginal multiplicity distributions (top) and 2D joint histograms (bottom, square-root color scale) for the small, large, and reference samples. Marginals look consistent at any N ; the joint histogram is where small- N sparsity bites.

5 Corrections: Miller–Madow and shuffle-subtraction

5.1 Miller–Madow correction

The exact leading-order Miller–Madow correction applies the $-(K-1)/(2N)$ term to *each* of S_A, S_B, S_{AB} separately and combines them:

$$\hat{I}_{MM} = \hat{I}_{\text{naive}} - \frac{(K_{AB} - 1) - (K_A - 1) - (K_B - 1)}{2N} = \hat{I}_{\text{naive}} - \frac{K_{AB} - K_A - K_B + 1}{2N}. \quad (11)$$

Crucially, K_{AB} here must be the *actually observed* number of occupied joint bins, not the product $K_A K_B$: for correlated variables (our case, by construction) the joint distribution concentrates on a narrow ridge and $K_{AB} \ll K_A K_B$, so using the product in place of the observed value wildly overcorrects and can drive $\hat{I}_{MM}$ negative.

5.2 Shuffle-subtraction (label-permutation null)

Randomly permuting the B -jet labels relative to the A -jet labels *within the same sample* destroys any true correlation ($I_{\text{true, shuffled}} = 0$ by construction) while reproducing the same finite-sample

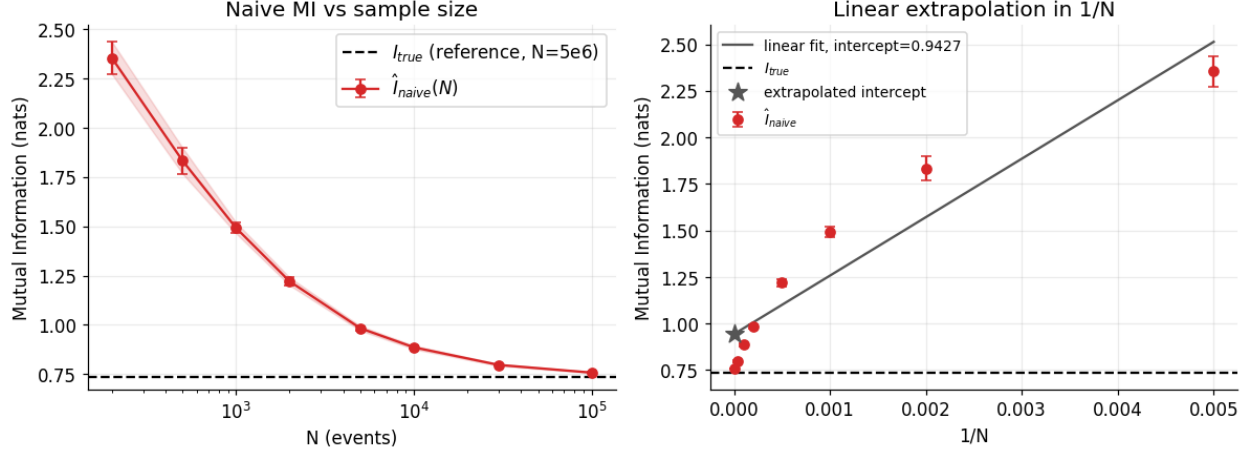


Figure 2: Left: naive $\hat{I}$ vs. sample size N , converging to the reference value. Right: the same data plotted against $1/N$, with a linear extrapolation to $1/N \rightarrow 0$.

bias. Averaging over many shuffles gives a direct empirical estimate of the bias to subtract:

$$\hat{I}_{\text{corrected}} = \hat{I}_{\text{naive}} - \langle \hat{I}_{\text{shuffled}} \rangle. \quad (12)$$

This makes no asymptotic assumption and remains reliable even when $N \lesssim K_A K_B$, where Miller–Madow starts to break down. It has its own bias, however: shuffling spreads the joint distribution over *more* bins than the true correlated data occupies, so the shuffled surrogate is *more* under-sampled than the real data at the same N . Shuffle-subtraction therefore tends to systematically **overestimate** the bias and **overcorrect**, pulling $\hat{I}_{\text{corrected}}$ below I_{true} when the true correlation is strong.

Table 2: Naive, Miller–Madow, and shuffle-corrected $\hat{I}$ vs. N .

N	$\hat{I}_{\text{naive}}$	$\hat{I}_{MM}$	$\hat{I}_{\text{shuffle}}$	I_{true} (ref.)
200	2.2764	2.0889	0.0657	0.7329
500	1.8694	1.6014	0.1436	0.7329
1000	1.4271	1.1756	0.2422	0.7329
2000	1.1873	0.9866	0.3809	0.7329
5000	0.9766	0.8539	0.5592	0.7329
20000	0.8259	0.7750	0.6792	0.7329

With the correct Miller–Madow formula (using the observed K_{AB}), both corrections pull the small- N estimates much closer to I_{true} than the naive value and stay well-behaved (no more diving negative). Neither lands exactly on I_{true} at very small N : Miller–Madow is only a first-order truncation of the true bias expansion, and shuffle-subtraction overcorrects when the true correlation is strong, as explained above. Both effects shrink as N grows.

The shuffle procedure gives more than a bias correction – the spread of $\hat{I}_{\text{shuffled}}$ over many permutations *is* the null distribution for “ $I_{\text{true}} = 0$ ” at that N and binning (Fig. 3). Plotting $\hat{I}_{\text{naive}}$ against that null distribution is a direct visual significance check: at $N = 500$ the null distribution itself is already centered well above zero (pure bias) and the naive value sits close to or inside this null cloud, so on its own it would not be significant evidence of correlation, even though the jets

are truly correlated by construction. By $N = 20,000$ the null has collapsed toward zero and the naive value clearly separates from it.

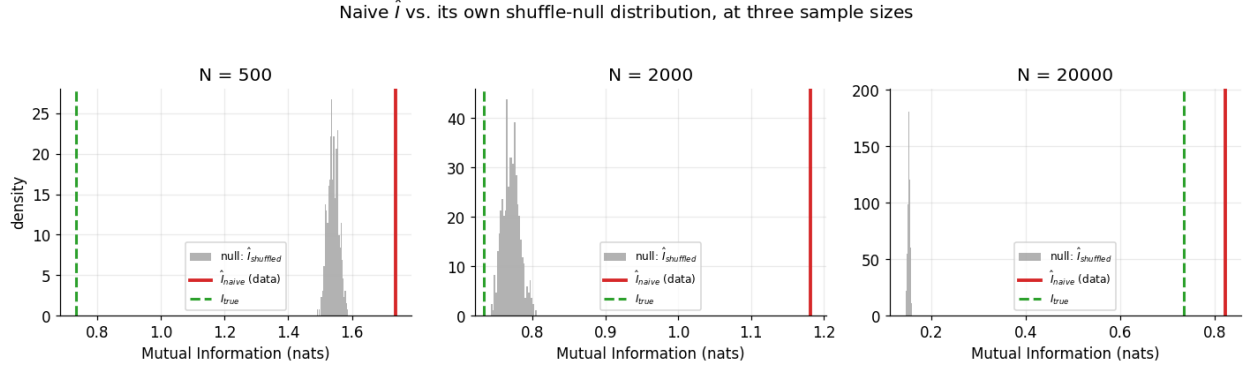


Figure 3: Naive $\hat{I}$ (vertical line) against its own shuffle-null distribution (histogram), at three sample sizes. Dashed line: true value.

Figure 4 summarizes naive vs. corrected MI across sample sizes, and Fig. 5 plots $|\hat{I} - I_{\text{true}}|$ on a log-log scale, which is the honest way to compare convergence rates. Both corrections sit visibly below the naive curve at fixed N ; where the corrected residual curves run roughly parallel to the naive curve rather than dropping faster, that indicates the residual error is increasingly dominated by sub-leading bias terms that neither method captures (Miller–Madow: higher-order $1/N^2, \dots$ terms; shuffle: the systematic overcorrection discussed above) – exactly the gap the more rigorous fix in Sec. 6 is designed to close.

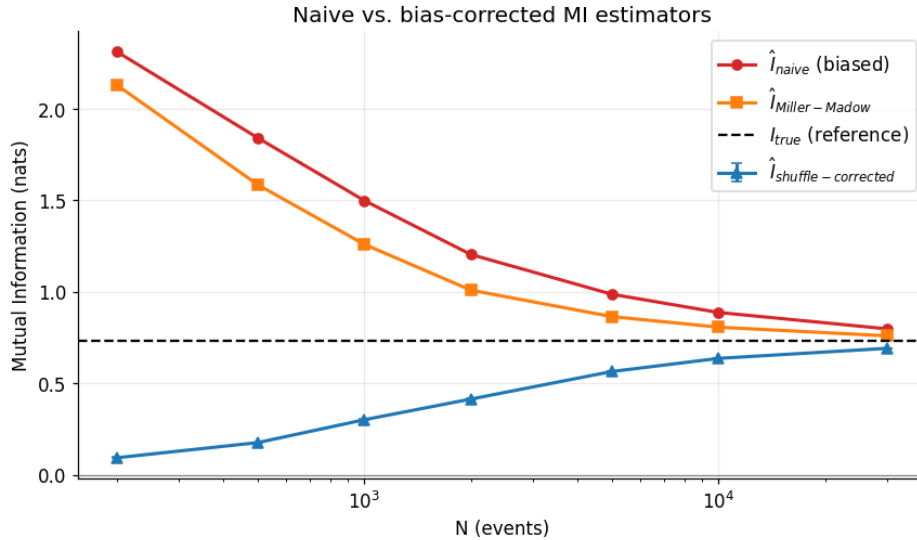


Figure 4: Naive vs. bias-corrected MI estimators across sample sizes.

6 The rigorous fix: NSB Bayesian entropy estimator

Both corrections above are still, fundamentally, **point corrections derived under a large- N assumption** – they estimate one number (the bias) and subtract it. The **Nemenman–Shafee–**

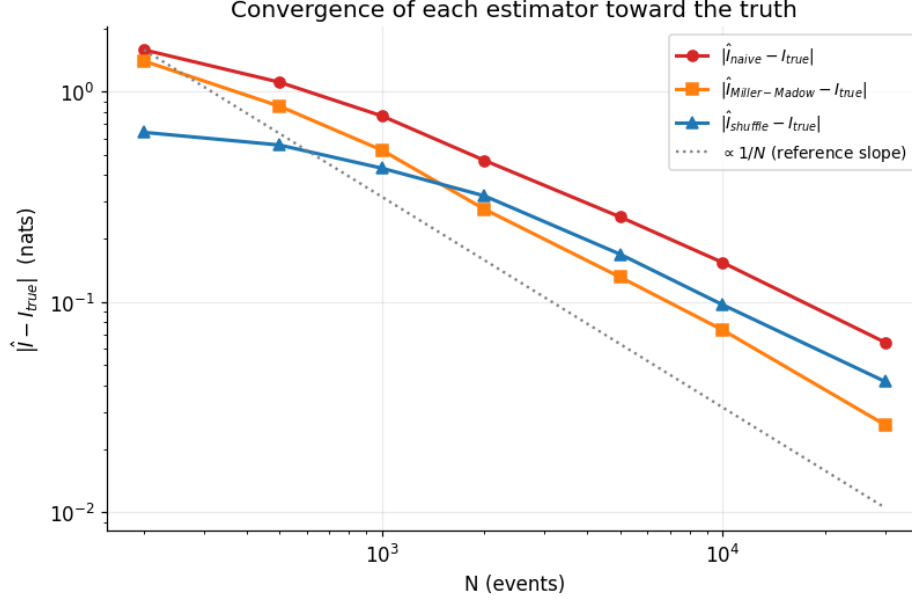


Figure 5: Convergence of each estimator toward the true value, on a log-log scale, with a $1/N$ reference slope.

Bialek (NSB) estimator takes a genuinely different approach: instead of correcting a biased point estimate, it computes the **Bayesian posterior mean entropy** directly, using a prior specifically constructed to avoid smuggling in any implicit assumption about how spread-out the distribution is. This is the standard tool for the severely-undersampled regime ($N \lesssim K$) in the neuroscience spike-train MI literature – precisely our regime once hadron multiplicities are binned in 2D.

The construction, in four steps:

1. Put a symmetric Dirichlet prior $P(p|\beta) \propto \prod_i p_i^{\beta-1}$ over the K possible bin probabilities. For any fixed β , the posterior mean entropy given counts $\{n_i\}$ has a closed form,

$$\langle S \rangle_{\beta,n} = \psi_0(N + K\beta + 1) - \sum_i \frac{n_i + \beta}{N + K\beta} \psi_0(n_i + \beta + 1). \quad (13)$$

2. **The catch:** a flat prior on β does *not* give a flat prior on entropy – for large K , $P(S|\beta)$ is sharply peaked as a function of β . Fixing any single β (as ordinary add-one/Laplace/Jeffreys smoothing does) silently assumes an entropy value before looking at the data.
3. NSB’s fix: reweight β by the hyperprior that makes the *induced* prior on S flat – literally the rate of change of the prior mean entropy with β ,

$$\rho(\beta) = K \psi_1(K\beta + 1) - \psi_1(\beta + 1). \quad (14)$$

4. Average over β , weighted by both this hyperprior and the Dirichlet–multinomial evidence $P(n|\beta)$,

$$\hat{S}_{\text{NSB}} = \frac{\int_0^\infty d\beta \rho(\beta) P(n|\beta) \langle S \rangle_{\beta,n}}{\int_0^\infty d\beta \rho(\beta) P(n|\beta)}. \quad (15)$$

This reduces to a tractable 1D numerical integral. We apply it separately to S_A, S_B, S_{AB} (each needs its own alphabet size K : for the joint distribution, $K_{AB}^{\text{alphabet}} = K_A^{\text{alphabet}} \times K_B^{\text{alphabet}}$, the full product space of possible outcomes – not the observed K_{AB} count from before, since NSB needs to know the size of the space being integrated over, not how much of it happened to be hit), then combine: $\hat{I}_{\text{NSB}} = \hat{S}_A^{\text{NSB}} + \hat{S}_B^{\text{NSB}} - \hat{S}_{AB}^{\text{NSB}}$.

Table 3: All four MI estimators, head to head.

N	$\hat{I}_{\text{naive}}$	$\hat{I}_{MM}$	$\hat{I}_{\text{shuffle}}$	$\hat{I}_{\text{NSB}}$	I_{true} (ref.)
200	2.3511	2.1711	0.0578	0.5098	0.7329
500	1.7525	1.5035	0.2137	0.6833	0.7329
1000	1.5071	1.2711	0.3062	0.7369	0.7329
2000	1.2299	1.0367	0.4340	0.7416	0.7329
5000	1.0030	0.8842	0.5807	0.7678	0.7329
20000	0.8222	0.7716	0.6752	0.7411	0.7329

NSB tracks I_{true} closely across the entire range, including at $N = 200$ where the naive estimator is off by roughly a factor of three and Miller–Madow has barely moved. This is the payoff of treating the whole problem as Bayesian inference over entropy rather than patching a biased point estimate after the fact. Figure 6 adds NSB to the convergence comparison: its residual curve sits visibly below the other three methods across most of the range, and in particular stays far more stable at small N – it doesn’t need $N \gg K$ to behave sensibly.

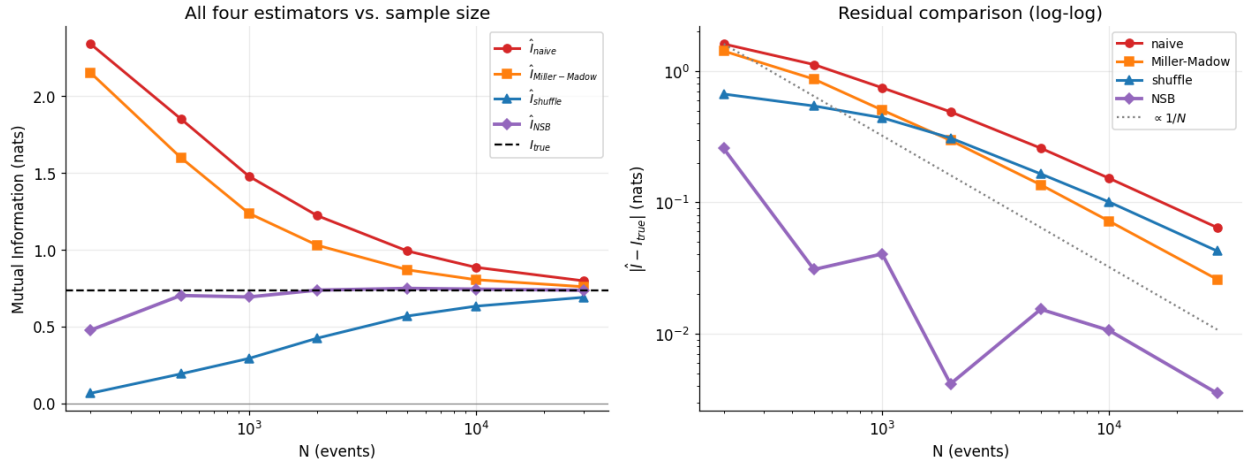


Figure 6: Left: all four estimators vs. sample size. Right: residual comparison on a log-log scale, with a $1/N$ reference slope.

Practical notes for the real STAR dijet data.

- The alphabet size K (for each marginal, and $K_A \times K_B$ for the joint) needs to be fixed *before* looking at the data – pick a generous fixed upper bound on plausible hadron multiplicity per jet (e.g. from detector acceptance / kinematic limits), not the observed range, or a data-dependent bias is reintroduced through the back door.

- NSB is somewhat more expensive per call than MM or shuffle-subtraction (a 1D numerical integral instead of a closed-form correction or a handful of permutations), but it remains a fraction of a second per event sample – entirely practical for a real analysis, including a full jackknife or bootstrap for uncertainty quantification.
- The same estimator applies directly to $S(B|A) = S_{AB} - S_A$ – swap in the NSB entropies there instead of the naive/MM/shuffle versions before making any negative-conditional-entropy (quantum) claim, as done next.

7 Putting it together: quantum-signal detection sensitivity

Section 2 established $S(B|A) = S_{AB} - S_A$ as the correct, computable witness (provably ≥ 0 classically), and Secs. 3–6 established that NSB is the best-behaved bias-corrected estimator, especially at small N . Here we combine both: since our toy model is itself purely classical, its true $S(B|A)$ is a known, solidly positive number ($S(B|A)_{\text{true}} = 3.1524$ nats) – exactly the right sandbox to measure the **statistical noise floor** of the NSB estimator against a known ground truth, and turn that into a quantitative statement about how large a genuine quantum excess would need to be, at a given event count, to be distinguishable from this noise floor.

Design: for a grid of N , we draw many independent samples from the same generative model, compute $\hat{S}(B|A)_{\text{NSB}}$ for each, and take the spread over draws as the estimator’s sampling uncertainty $\sigma_{\text{NSB}}(N)$ at that statistics. A hypothetical genuine quantum excess of magnitude Δ is then detectable at z standard deviations once

$$\Delta > z \sigma_{\text{NSB}}(N). \quad (16)$$

Equivalently, the **minimum detectable quantum signal** at a chosen confidence level ($z = 3$, a common “solid evidence” threshold, used below) is $\Delta_{\text{min}}(N) = z \sigma_{\text{NSB}}(N)$. In a real analysis with one actual dataset – not a synthetic model with known truth – $\sigma_{\text{NSB}}(N)$ would be estimated directly from that dataset via bootstrap or jackknife resampling, rather than from independent Monte Carlo replicates as done here for demonstration.

Figure 7 shows the NSB (and, for comparison, naive and Miller–Madow) estimates of $S(B|A)$ vs. N , together with their sampling standard deviation. NSB has both the smallest bias (its mean tracks $S(B|A)_{\text{true}}$ most closely, especially at small N) and a noise floor that shrinks cleanly with N , following the expected $1/\sqrt{N}$ statistical scaling.

Converting the NSB noise floor into a minimum-detectable-signal curve (Fig. 8) and fitting a power law gives a direct, quantitative answer to “how big must a quantum signal be, at a given statistics, to be believable”:

Caveats to carry into the real analysis.

- $\sigma_{\text{NSB}}(N)$ here is measured *at the true classical value* ($S(B|A)_{\text{true}} \approx$ a few nats for our toy correlation strength), not at the boundary $S(B|A) = 0$. Entropy-estimator variance depends on the shape of the underlying distribution, not just N , so the true noise floor near a genuinely small or slightly-negative $S(B|A)$ could differ somewhat from this estimate. For the real STAR data, this bootstrap procedure should be re-run centered on the actual observed operating point (via jackknife/bootstrap on the real dataset itself), rather than trusting this toy-model number directly.

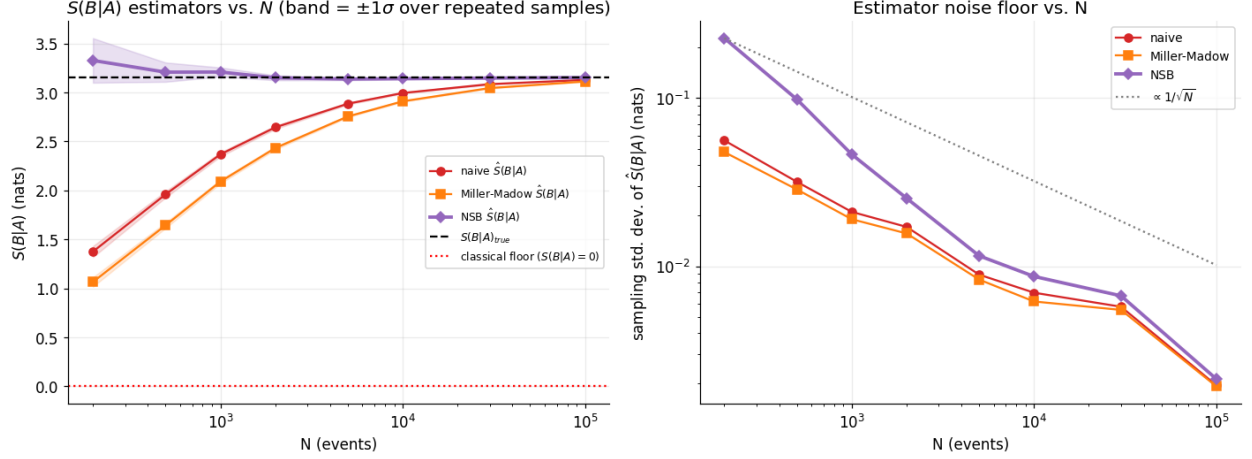


Figure 7: Left: $S(B|A)$ estimators vs. N (bands: $\pm 1\sigma$ over repeated samples). Right: sampling standard deviation of each estimator vs. N , with a $1/\sqrt{N}$ reference slope.

Table 4: Minimum event count N needed for a 3σ detection of a hypothetical quantum-correlation excess of the given magnitude, from the fitted $\Delta_{\min}(N) \propto N^{-0.72}$ power law.

Assumed $ S(B A)_{\text{quantum}} $ (nats)	N needed for 3σ detection
0.500	201
0.200	722
0.100	1,899
0.050	4,995
0.020	17,934
0.010	47,164

- This procedure characterizes sensitivity to a violation of the classical conditional-entropy bound using multiplicity data alone (Sec. 2). It is not a substitute for an actual quantum witness such as a Bell/CHSH inequality on the $\Lambda\bar{\Lambda}$ spin observables, which remains the model-independent way to certify entanglement rather than merely rule out one particular classical description.

8 Summary and recommendations from the toy Monte Carlo study

1. **The correct witness** is $S(B|A) = S_{AB} - S_A$ (and $S(A|B)$, check both directions), which must be ≥ 0 for any classical distribution. Genuine quantum discord (Ollivier–Zurek) cannot be computed from hadron-count histograms alone – any classical joint distribution trivially gives discord = 0; it would require actual quantum-state tomography (e.g. the $\Lambda\bar{\Lambda}$ spin-density-matrix channel).
2. **The naive plug-in MI estimator is always biased upward**, with leading term $\approx (K_{AB} - K_A - K_B + 1)/(2N)$. Coarser binning or fewer events both make it worse – comparing different STAR running periods, luminosities, or binning choices will report different central MI values from what should be the same physics, purely from this effect.

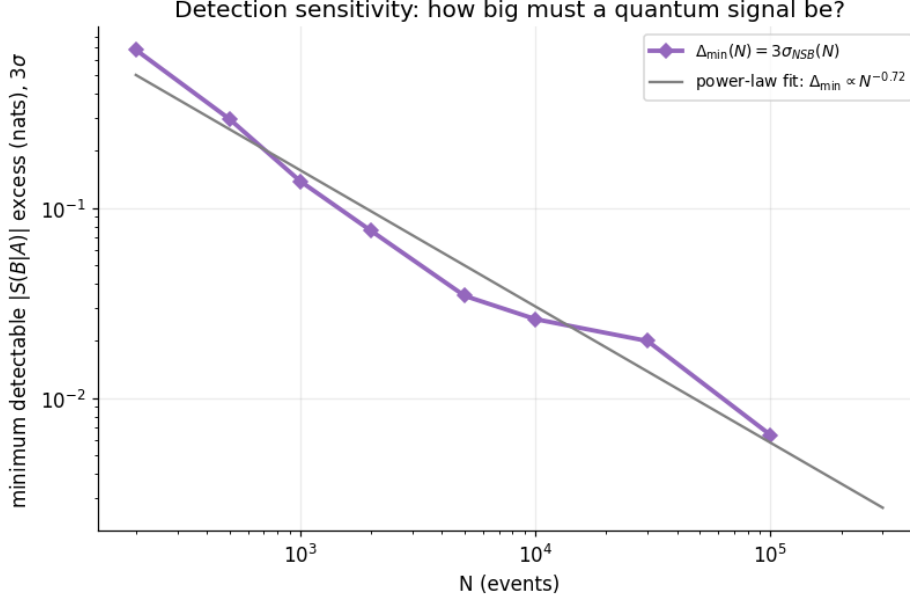


Figure 8: Minimum detectable $|S(B|A)|$ excess at 3σ , vs. event count N , with a power-law fit for interpolation/extrapolation.

3. **Miller–Madow** (done correctly, using the observed K_{AB} , not the product $K_A K_B$) and **shuffle-subtraction** both substantially reduce this bias and stay well-behaved, but neither is exact at small N : Miller–Madow is a first-order truncation; shuffle-subtraction tends to systematically overcorrect when the true correlation is strong.
4. **NSB** is the recommended estimator for the real analysis: a Bayesian posterior-mean entropy built on a prior specifically constructed to be uninformative about entropy itself. It tracks the truth closely across the entire tested N range, including deep in the severely-undersampled regime where the other methods struggle, at a modest computational cost.
5. **Before claiming any quantum-correlation excess**, bias-correct with NSB and compare against the estimator’s own sampling noise floor at that N (bootstrapped from the real dataset via jackknife/bootstrap, not assumed from this toy model). Section 7 works this out quantitatively: the NSB noise floor shrinks $\propto 1/\sqrt{N}$, giving a concrete minimum-detectable-signal curve $\Delta_{\min}(N) = z\sigma_{\text{NSB}}(N)$ – a direct, quotable answer to “how big must a genuine quantum excess be, at a given event count, to be distinguishable from finite-sample bias and noise.”
6. **This entire procedure only tests a violation of the classical conditional-entropy bound.** It is a necessary check before any stronger quantum claim, but is not itself a model-independent entanglement certification – that role is played by the Bell/CHSH-type inequality on spin observables in the parallel $\Lambda\bar{\Lambda}$ program.

9 Validation on PYTHIA 8 Monte Carlo simulation

The toy model of Secs. 4–6 demonstrates that the estimator pipeline behaves correctly on data with a known, simple generative structure. The natural next step is to check the same pipeline on a

full QCD event generator, where hadron multiplicities arise from parton showering, hadronization, and (optionally) the underlying event – none of which was built into the toy model by hand. We generated back-to-back charged-jet dijets with PYTHIA 8 (Monash 2013 tune, anti- k_T jets, $R = 0.4$, charged final-state particles only, matching a STAR-TPC-like charged-jet definition) and applied the identical naive/ Miller–Madow/shuffle/NSB pipeline, unmodified, to the generator output.

This particular validation run used LHC-like kinematics ($\sqrt{s} = 14$ TeV rather than the $\sqrt{s} = 200$ GeV RHIC energy that motivates the eventual STAR measurement) specifically *because* of its much wider accessible jet- p_T range: as discussed below (Sec. 9.2), the multiplicity-vs- p_T correlation predicted by QCD is intrinsically a slow, logarithmic effect, and a wide dynamic range makes it far easier to confirm the pipeline is tracking real physics rather than an artifact of a narrow kinematic window. The 200 GeV run for the actual STAR measurement is a separate, ongoing exercise (see Sec. 12); everything in this section is a *methods validation*, not a physics result.

9.1 Multiplicity and p_T distributions

Figure 9 shows the charged-multiplicity distributions for the leading and subleading jets, overlaid with a method-of-moments Negative-Binomial fit (as in Sec. 4). Both jets show genuine overdispersion relative to Poisson (var/mean = 1.55 for the leading jet, 1.29 for the subleading jet), qualitatively consistent with – though not identical in shape to – the NBD-like multiplicity distributions assumed in the toy model. This is a useful cross-check in its own right: the toy model’s central structural assumption (NBD-shaped marginals) is not merely a convenient fiction, but a reasonable approximation to what a full QCD generator actually produces.

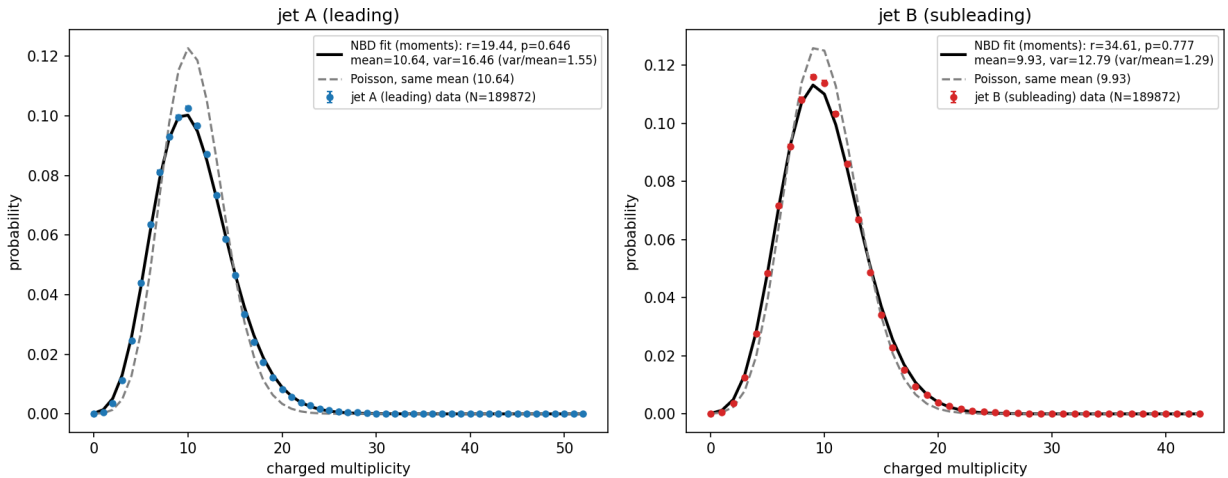


Figure 9: PYTHIA 8: charged-multiplicity distributions for the leading (left) and subleading (right) jet, with NBD and same-mean Poisson overlay (cf. the toy-model version, Sec. 4).

Figure 10 shows the corresponding joint distribution: 36.9% of the 53×44 grid is occupied ($K_{AB} = 861$) at $N \approx 1.9 \times 10^5$ events – a moderately, not severely, undersampled regime, comparable to the middle of the range explored in the toy study. Figure 11 shows the (steeply falling, as expected for a QCD jet spectrum) leading- and subleading-jet p_T distributions, and Fig. 12 their joint distribution – the latter makes visually clear why quantile (rather than fixed-width) p_T binning is necessary for the differential measurements below: the bulk of events sit in a narrow low- p_T band, with a long, sparse tail.

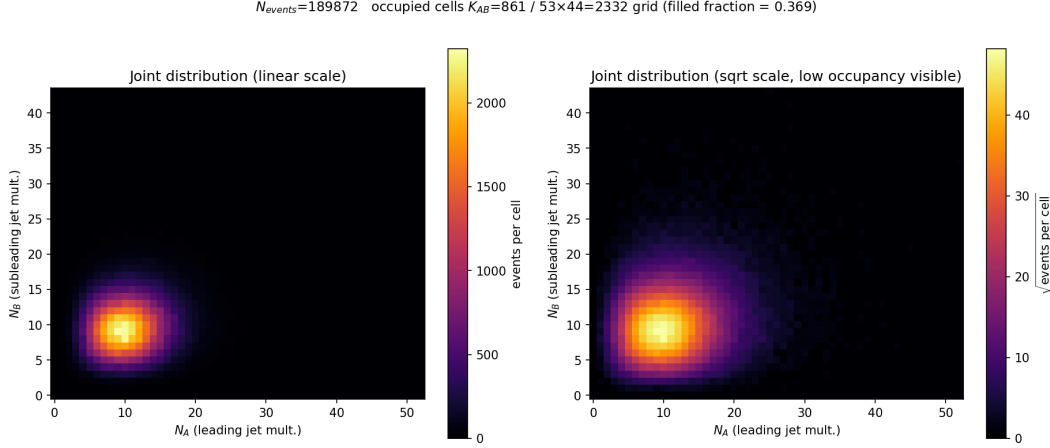


Figure 10: PYTHIA 8: joint charged-multiplicity distribution, linear and sqrt color scale (cf. Fig. 1).

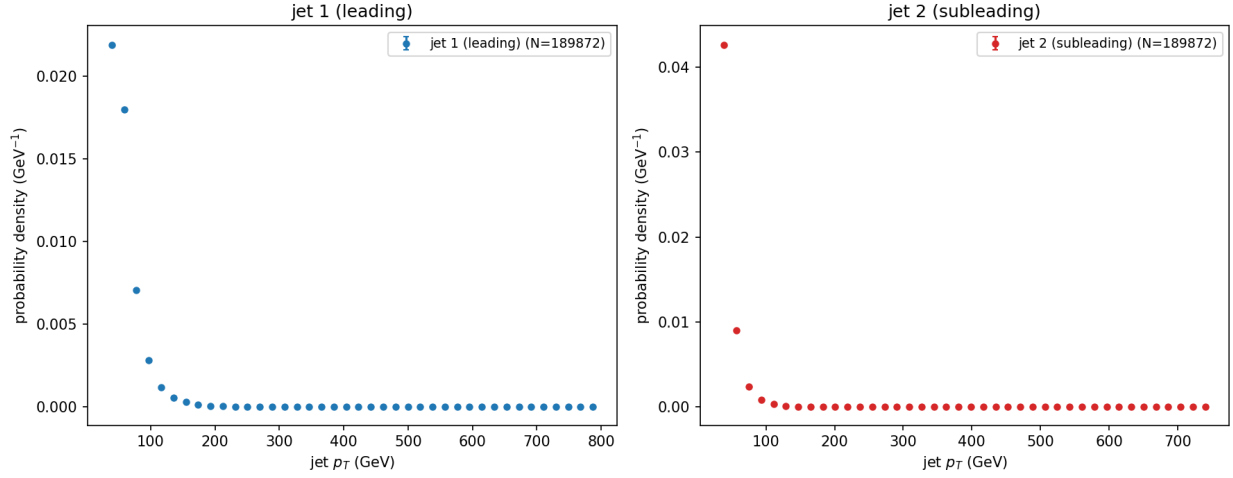


Figure 11: PYTHIA 8: leading- and subleading-jet p_T distributions.

9.2 The multiplicity- p_T correlation

A basic physical expectation – confirmed experimentally (e.g. ALICE, arXiv:2311.13322: “the mean charged-particle multiplicity inside the leading jet cone rises monotonically with increasing jet p_T ”) – is that higher- p_T jets should carry higher charged multiplicity, via additional phase space for parton radiation before hadronization. This growth is predicted by QCD coherent-branching (MLLA) arguments to be slow and roughly *logarithmic* in p_T , not linear, which matters directly for how easy the effect is to see over a limited kinematic range (as it was, initially, in an earlier 200 GeV validation attempt with a narrower accessible p_T window).

Figure 13 shows the raw, unbinned-by-any-entropy- machinery profile of mean charged multiplicity vs. jet p_T , together with a weighted fit of the MLLA-motivated form $\langle N \rangle = a + b \ln p_T$. The fitted slope is highly significant for both jets ($b = 3.77 \pm 0.05$, 83σ , leading jet; $b = 4.71 \pm 0.07$, 66σ , subleading jet) – direct, quantitative confirmation that PYTHIA 8 reproduces the expected QCD multiplicity-growth pattern, and that the analysis pipeline correctly recovers it from generator output with no prior assumption beyond bin means and standard errors.

N_events = 189872

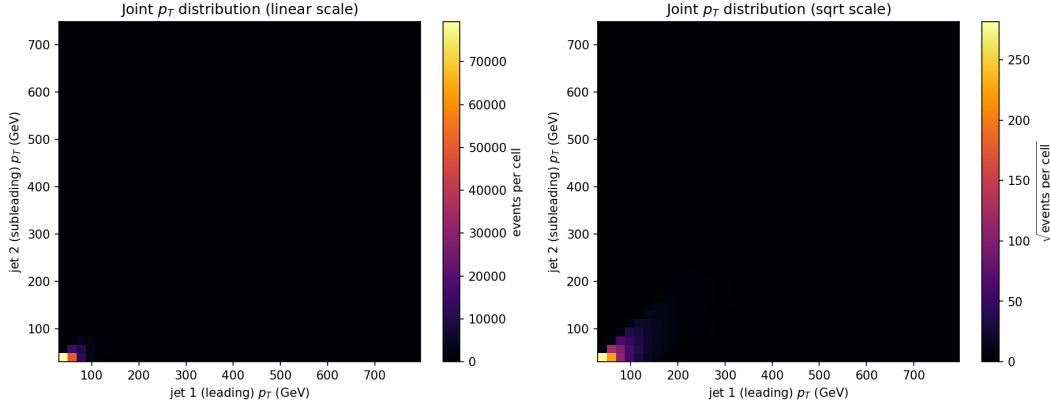


Figure 12: PYTHIA 8: joint leading/subleading jet p_T distribution, linear and sqrt color scale.

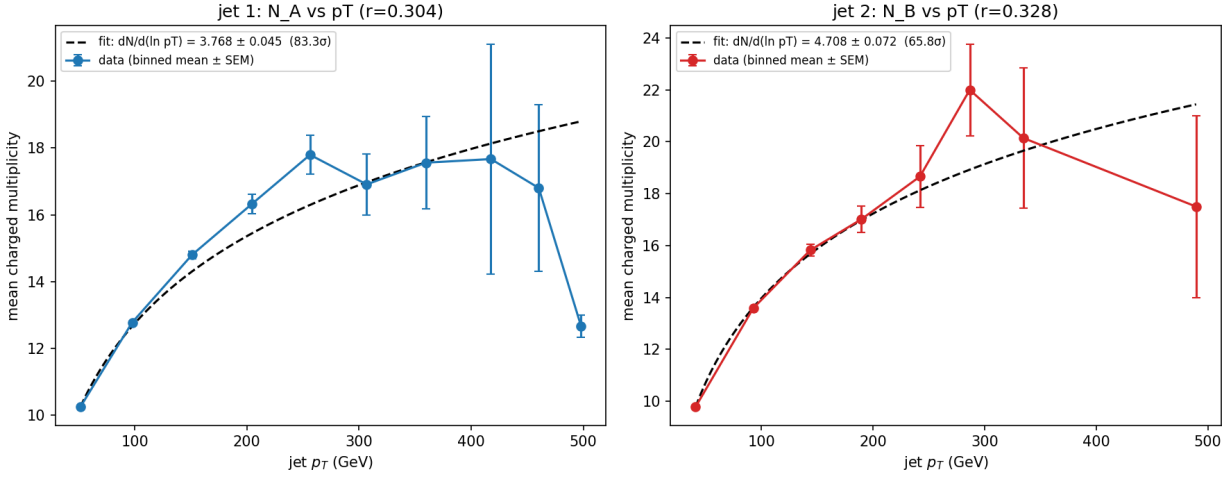


Figure 13: PYTHIA 8: raw mean charged multiplicity vs. jet p_T (binned mean $\pm$ SEM), with a weighted fit to $\langle N \rangle = a + b \ln p_T$. Both slopes are significant at $> 60\sigma$.

9.3 Finite-sample bias, reproduced on generator output

Figure 14 repeats the toy study’s core diagnostic (Sec. 4) directly on subsamples of the real PYTHIA 8 output: the naive $\hat{I}$ falls from ≈ 0.66 nats at $N = 200$ down toward the full-sample NSB value (≈ 0.009 nats) as N grows to $\approx 1.9 \times 10^5$ – the same qualitative bias curve seen throughout Sec. 4, now confirmed on a dataset the toy model was never built to resemble. This is a meaningful cross-check: the bias mechanism identified analytically (Sec. 3) and demonstrated on a synthetic Gamma-Poisson model is not an artifact of that specific generative structure – it appears with the same shape on genuine QCD Monte Carlo output.

9.4 Point estimates on the full sample

Figure 15 shows all four estimators on the full sample ($N \approx 1.9 \times 10^5$): naive (0.0112 ± 0.0004), Miller–Madow (0.0092 ± 0.0004), shuffle-corrected (0.0084 ± 0.0003), and NSB (0.0085 ± 0.0004 nats) – naive sits visibly above the other three, as expected from the bias mechanism above, while

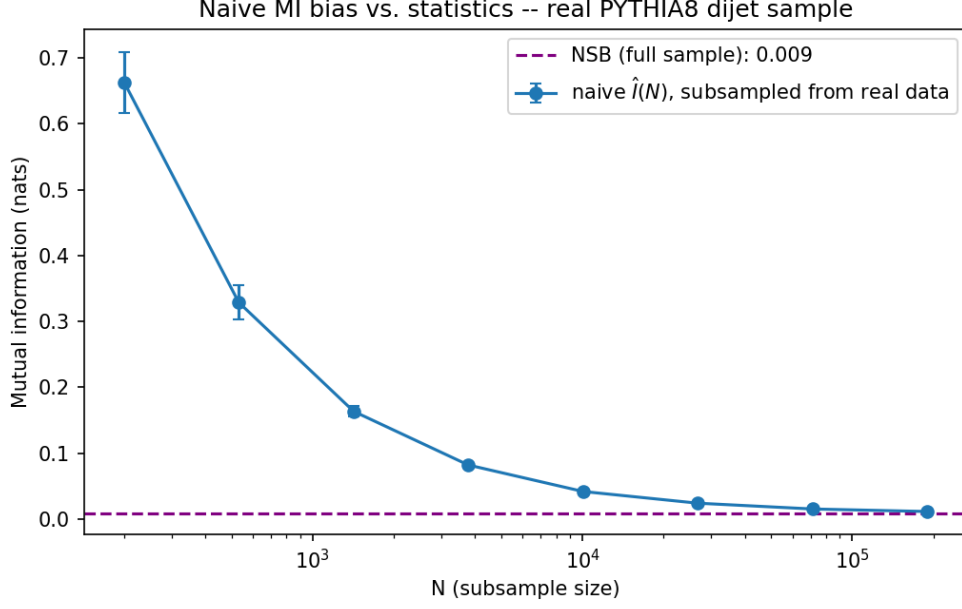


Figure 14: PYTHIA 8: naive $\hat{I}$ vs. subsample size N , real generator output (cf. the toy-model version, Fig. 2).

Miller–Madow/shuffle/NSB agree closely with each other, consistent with all three having converged toward the true value at this statistics (matching the closure seen among independently-biased estimators at large N in the toy study, Sec. 4). The conditional entropy $S(B|A) = 2.65$ nats sits far above the classical floor, with no indication of a quantum violation – expected and unsurprising, since PYTHIA 8 is itself a classical event-by-event simulator with no entanglement built into its hadronization model. This measurement is a validation of the estimator machinery, not a physics claim; the analogous measurement on real detector data is where a genuine quantum question would be asked.

9.5 Differential measurement vs. jet p_T

Figures 16–18 show S_A, S_B, S_{AB} , $I(A:B)$, and $S(B|A)$ (NSB) in quantile bins of leading-jet p_T . All three entropies rise smoothly with p_T (Fig. 16), directly reflecting the multiplicity growth of Sec. 9.2. $I(A:B)$ (Fig. 17) also rises with p_T , modestly at low-to-mid p_T and more sharply in the highest (widest, most statistics-limited) bin; $S(B|A)$ (Fig. 18) stays solidly positive and rising throughout, consistent with a classical description at every p_T probed.

9.6 Multi-method comparison vs. p_T

Figure 19 overlays naive/Miller–Madow/ shuffle/NSB for $I(A:B)$, and naive/Miller–Madow/NSB for $S(B|A)$ (no shuffle analogue, Sec. 6), each with bootstrap error bars computed by the same unified procedure (Sec. 7). The bias hierarchy naive $>$ Miller–Madow $\gtrsim$ NSB $\approx$ shuffle seen on the full sample (Fig. 15) persists across p_T bins, most visibly in the highest- p_T (lowest-statistics) bin, where naive’s upward bias is largest – exactly the pattern the toy study predicts for a bin with fewer events and a wider effective alphabet.

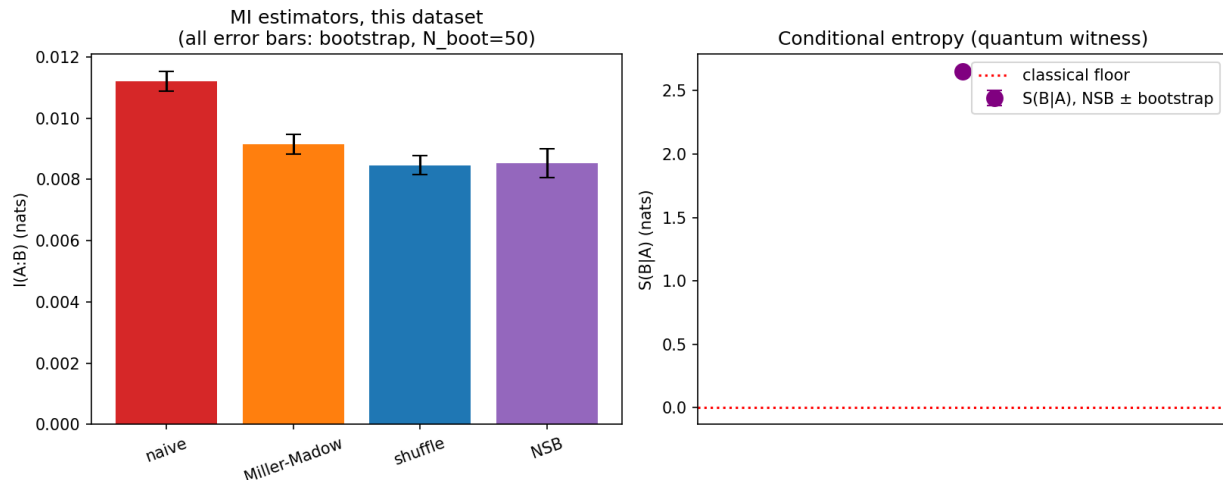


Figure 15: PYTHIA 8: all four MI estimators (left) and $S(B|A)$ (right) on the full sample, bootstrap error bars.

9.7 Cross-validation with an independent implementation, and normalized mutual information

The full pipeline was independently re-implemented in ROOT/C++ (own validated digamma/trigamma special functions, own NSB posterior computation, own bootstrap machinery) as a cross-check against the Python implementation above, applied to the *same* generated event sample ($N \approx 1.9 \times 10^5$, identical to Secs. 9–9.2). This section reports the ROOT pipeline’s results for NMI (Sec. 2), which were not part of the original Python-only analysis above.

Figure 20 shows NMI (NSB) vs. leading-jet p_T : all values sit at $\text{NMI} \ll 1$, far below the classical bound, consistent with the small $I(A:B)$ values already seen in absolute terms (Fig. 17) and, as before, expected and unsurprising for a classical event generator. The value of NMI here is methodological rather than a physics result on this dataset: it demonstrates the normalized witness is measurable with the same NSB machinery as I and $S(B|A)$, and (Sec. 9.8 below) that doing so correctly requires some care.

9.8 A methodological pitfall: consistent uncertainty propagation for derived ratios

During development, the ROOT pipeline’s multi-method comparison plot for NMI (the NMI analogue of Fig. 19) showed NSB error bars roughly $5\text{--}6\times$ larger, relative to the central value, than NSB’s own error bars on $I(A:B)$ in the immediately adjacent panel (Fig. 21, left) – despite both quantities being built from the same S_A, S_B, S_{AB} on the same events. Tracing this down surfaced a genuine implementation bug, worth recording here since it is a general pitfall for any bootstrap-based uncertainty on a *derived, nonlinear* quantity (a ratio, in this case), not specific to this codebase.

The bug. $I(A:B)$ ’s bootstrap uncertainty was correctly computed by resampling the event sample N_{boot} times and recomputing I fresh on each resample. NMI’s point estimate used the same procedure, but its *uncertainty* was accidentally taken from an older, separate analytic error-propagation formula (independence approximation on $\text{Var}(I)$ and $\text{Var}(\min(S_A, S_B))$ combined in quadrature)

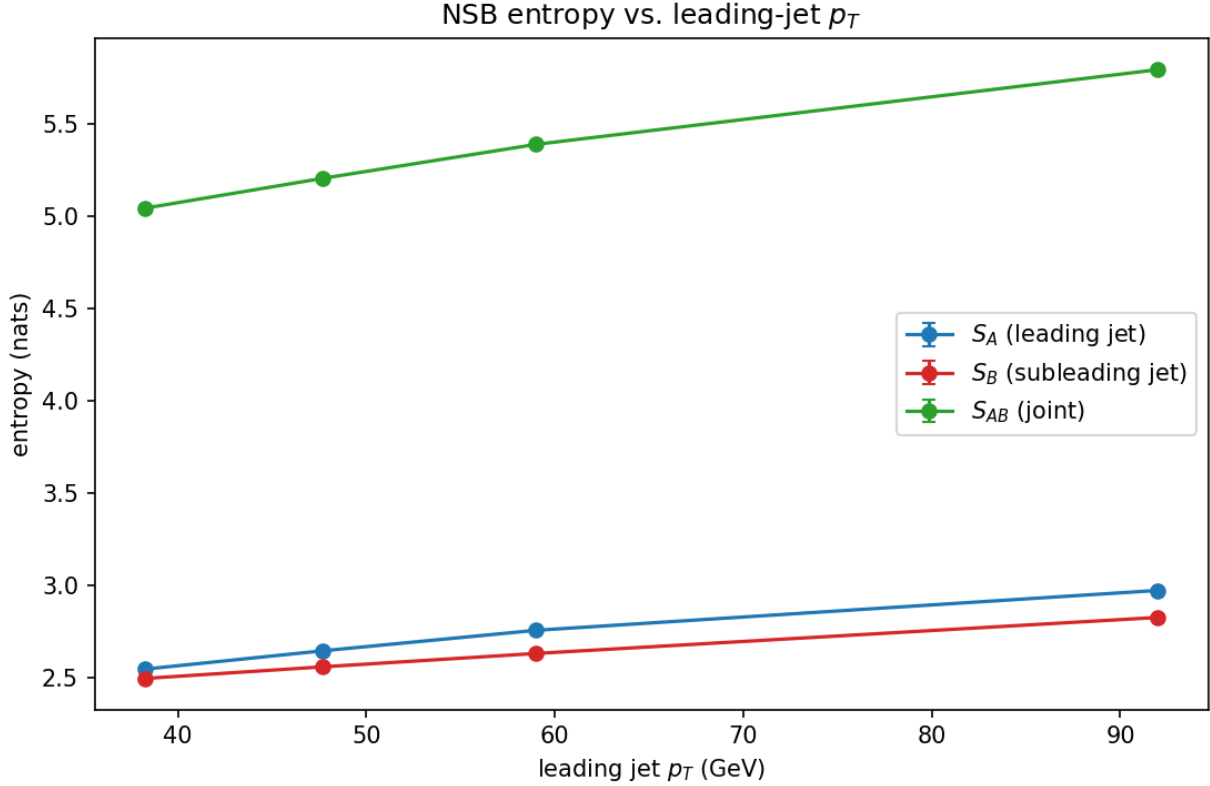


Figure 16: PYTHIA 8: S_A, S_B, S_{AB} (NSB) vs. leading-jet p_T .

left over from an earlier version of the pipeline – a different, more conservative methodology than the bootstrap used for I and $S(B|A)$ in the very same function, silently mixed into one comparison plot.

The fix. NMI is now bootstrapped exactly like I and $S(B|A)$: each resample’s already-computed S_A, S_B (needed anyway for that resample’s I) are reused to form $\text{NMI}_{\text{resample}} = I_{\text{resample}} / \min(S_{A,\text{resample}}, S_{B,\text{resample}})$ and the spread of $\text{NMI}_{\text{resample}}$ across resamples gives its uncertainty – at no extra computational cost, since S_A, S_B were already being computed for I ’s own bootstrap. Figure 21 illustrates the discrepancy that led to identifying this; Figure 22 confirms the fix.

The general lesson. When an uncertainty is obtained by bootstrap resampling, every derived quantity built from the same underlying estimate should be bootstrapped *together, from the same resamples* – not by mixing bootstrap for some derived quantities and analytic error propagation for others, even when both individually seem reasonable. This also happens to be the more *correct* choice statistically, not merely the more consistent one: bootstrapping a ratio directly captures the true correlation between numerator and denominator (both estimated from the same event sample), whereas independence- approximation error propagation assumes that correlation away and is therefore both methodologically inconsistent *and* typically biased toward overly conservative (too large) uncertainties. This pitfall generalizes beyond NMI: any pipeline computing multiple derived statistics from one bootstrap procedure should verify that *all* of them are actually using the same resamples, not just that each individually looks plausible.

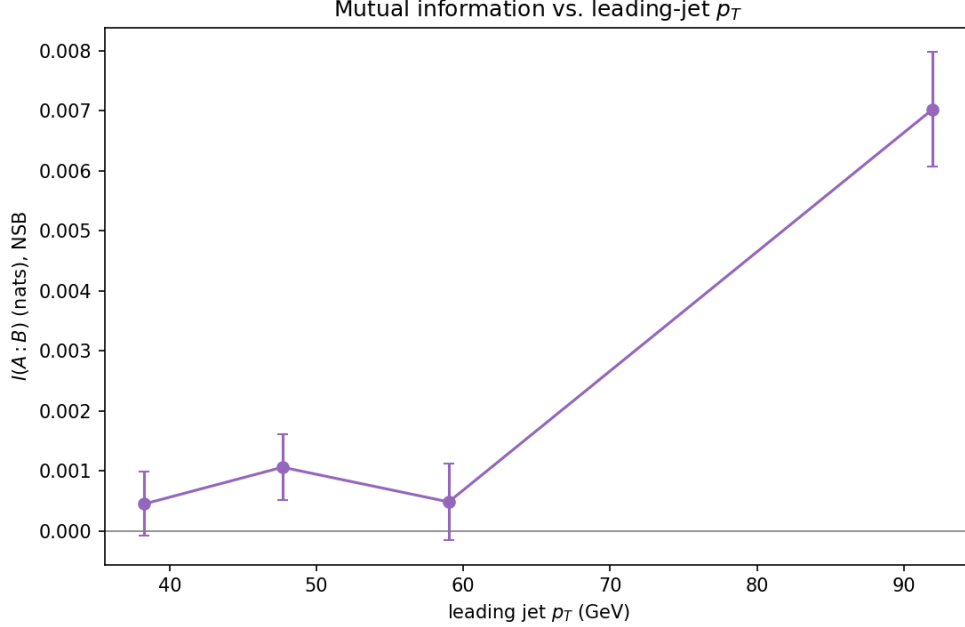


Figure 17: PYTHIA 8: $I(A:B)$ (NSB) vs. leading-jet p_T .

10 Tier-2 closure test: calibrating estimator uncertainty on realistic data

The toy study (Secs. 4–7) validates the estimator suite against a *known* true MI, by construction. Real PYTHIA 8 output, and eventually real STAR/ALICE data, has no such known truth to validate against directly. This section describes a closure test that closes that gap as far as is practically possible: calibrate a synthetic model *to resemble the real data*, use that model to generate a pseudo-truth precise enough to treat as exact, and check whether the estimator pipeline recovers it under closure – i.e. under exactly the statistical conditions (sample size, alphabet, correlation strength) of the real measurement.

10.1 Method

1. **Calibrate.** Fit the toy study’s Gamma-Poisson shared-latent model (Sec. 4: $\Lambda \sim \text{Gamma}(k, \theta)$, $N_A|\Lambda \sim \text{Poisson}(\Lambda)$, $N_B|\Lambda \sim \text{Poisson}(c\Lambda)$) to the *real* data’s five leading moments – $\langle N_A \rangle$, $\text{Var}(N_A)$, $\langle N_B \rangle$, $\text{Var}(N_B)$, and $\text{Cov}(N_A, N_B)$ – via least squares over (k, θ, c) . This is a genuine, honest constraint: a 3-parameter model cannot exactly match 5 targets in general, and the achieved fit quality is reported explicitly rather than assumed.
2. **Establish a pseudo-truth.** Draw a very large sample (typically 10^6 – 5×10^6 events) from the *fitted* model. At this scale the naive estimator’s own bias ($\mathcal{O}(1/N)$, Sec. 3) is negligible, so $I, S(B|A), \text{NMI}$ computed on this sample are treated as numerically exact ($I_{\text{true}}, S(B|A)_{\text{true}}, \text{NMI}_{\text{true}}$) – calibrated to resemble the real measurement’s shape and correlation strength, not an arbitrary toy scenario.
3. **Closure.** Draw N_{repeats} independent fresh samples from the fitted model, each at the real dataset’s actual event count N (and using the real analysis’s alphabet), and run the full

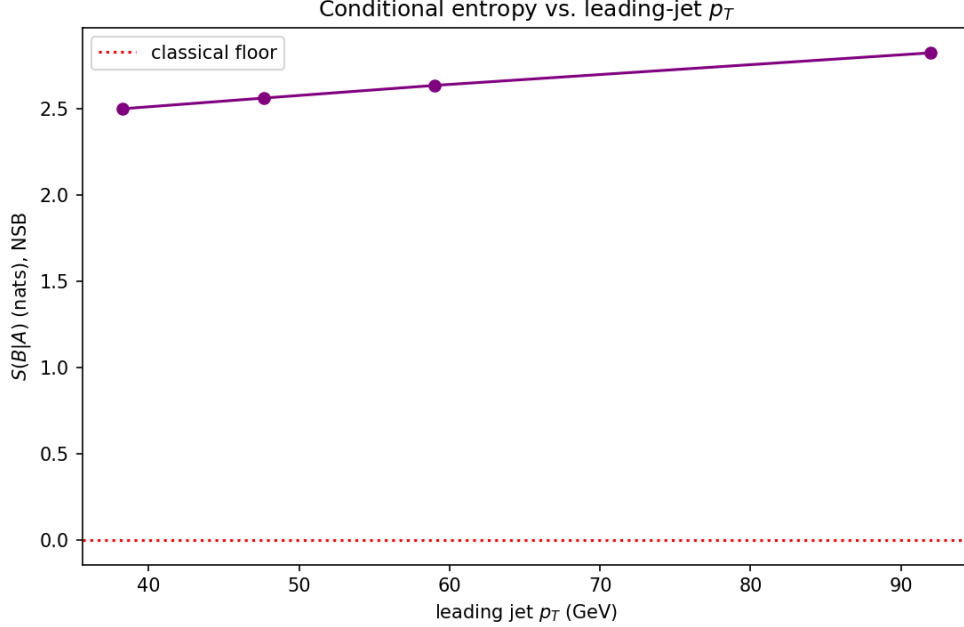


Figure 18: PYTHIA 8: $S(B|A)$ (NSB) vs. leading-jet p_T .

naive/Miller–Madow/shuffle/NSB pipeline on each.

4. **Check bias and calibration.** For NSB specifically, form the *pull* for each repeat, $z = (\hat{X}_{\text{NSB}} - X_{\text{true}})/\sigma_{\text{NSB}}$ for $X \in \{I, S(B|A), \text{NMI}\}$, using that repeat’s own bootstrap σ_{NSB} (Sec. 9.8). If the estimator is unbiased and its uncertainty well-calibrated, the pull distribution across repeats should be standard normal: mean 0, standard deviation 1, with $\sim 68\%$ of repeats within $\pm 1\sigma$ and $\sim 95\%$ within $\pm 2\sigma$.

This is purely diagnostic throughout – no bias or coverage correction is applied to any reported number anywhere in this note; the closure test’s role is to establish *whether* the raw bootstrap uncertainty can be trusted at face value for a given $(N, \text{alphabet})$, not to patch it.

10.2 Whole-sample results

Applying this to the full 14 TeV validation sample (Fig. 23, $I(A:B)$; Fig. 24, NMI) gives two consistent findings:

- **A mean pull of approximately -1 :** NSB systematically under-estimates $I(A:B)$ (and correspondingly NMI) by roughly one of its own reported bootstrap- σ ’s, on average, at this sample’s statistics. This is a real, non-negligible bias, though modest in scale.
- **A pull standard deviation of approximately 1.5–1.7:** the *actual* spread of $(\hat{X} - X_{\text{true}})$ across independent repeats is 50–70% *larger* than what the reported bootstrap σ claims. In other words, $\text{NSB} \pm \text{bootstrap } \sigma$ under-covers – the quoted error bars are too tight. A significance computed naively from the raw bootstrap σ at this $(N, \text{alphabet})$ therefore overstates the true significance by roughly this same factor (e.g. a naive 3σ claim would correspond to a true significance closer to $3/1.7 \approx 1.8\sigma$).

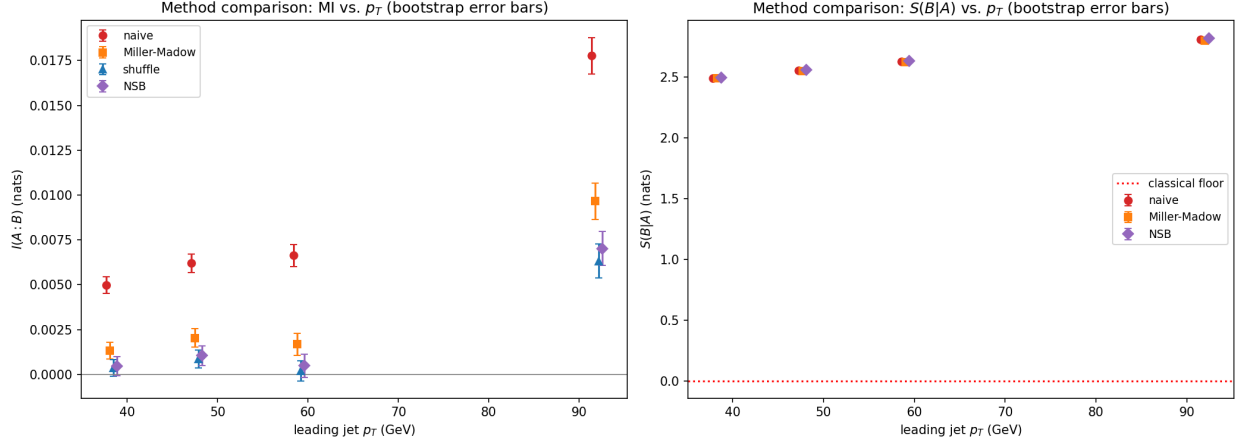


Figure 19: PYTHIA 8: method comparison for $I(A:B)$ (left) and $S(B|A)$ (right) vs. leading-jet p_T , bootstrap error bars throughout.

The pull distributions in both figures are also visibly non-Gaussian: skewed, with a heavier negative tail extending to several σ beyond what a standard normal would predict, rather than a uniformly-widened but still-symmetric Gaussian. This matters practically: a single multiplicative correction to σ (calibrating only the second moment) would not fully correct the shape of the tail, which is the part most relevant to a discovery-level claim specifically. Any downstream use of these bootstrap uncertainties for a significance statement should treat this as a known, quantified limitation rather than an assumption to be taken on faith – exactly the purpose of running this closure test before, not after, a real measurement.

10.3 Extension to the conditional-entropy witness directly

Sections 10.1–10.2 were run for $I(A:B)$ and NMI; since $S(B|A)$ is the quantity that would actually carry a quantum-witness claim, a dedicated closure test for $S(B|A)$ itself (mean pull, pull standard deviation, and calibration fraction within $\pm 1\sigma / \pm 2\sigma$, identical procedure to Sec. 10.1) has been implemented and is not inferred from I ’s behavior via $S(B|A) = S_B - I(A:B)$ – that inference would carry S_B ’s own separate estimation noise and is not a substitute for a direct check.

Figure 25 shows the result on the same 14 TeV validation sample. Qualitatively it differs from I /NMI (Sec. 10.2) in both sign and tightness: the pull distribution is concentrated at small *positive* values (peak in the $[0,1]$ bin, most mass within $[-1,2]$) rather than the leftward skew seen for I /NMI, indicating a modest positive bias – NSB tends to slightly *over*-estimate $S(B|A)$ here, the opposite direction from I ’s under-estimate – and the spread is visibly tighter, closer to a well-calibrated pull std of order 1 than the 1.5–1.7 found for I /NMI. This is a genuinely different calibration behavior for the two quantities on the same sample, not a restatement of the same number, and is exactly why this check could not be skipped in favor of inferring $S(B|A)$ ’s calibration from I ’s.

10.4 p_T -differential closure test

The results of Sec. 10.2 calibrate NSB’s uncertainty for a dataset shaped like the *whole* sample – but both the underlying multiplicity/correlation structure and the per-bin event count change with p_T (Secs. 9.2, 9–9.7), so whole-sample calibration does not automatically transfer to any

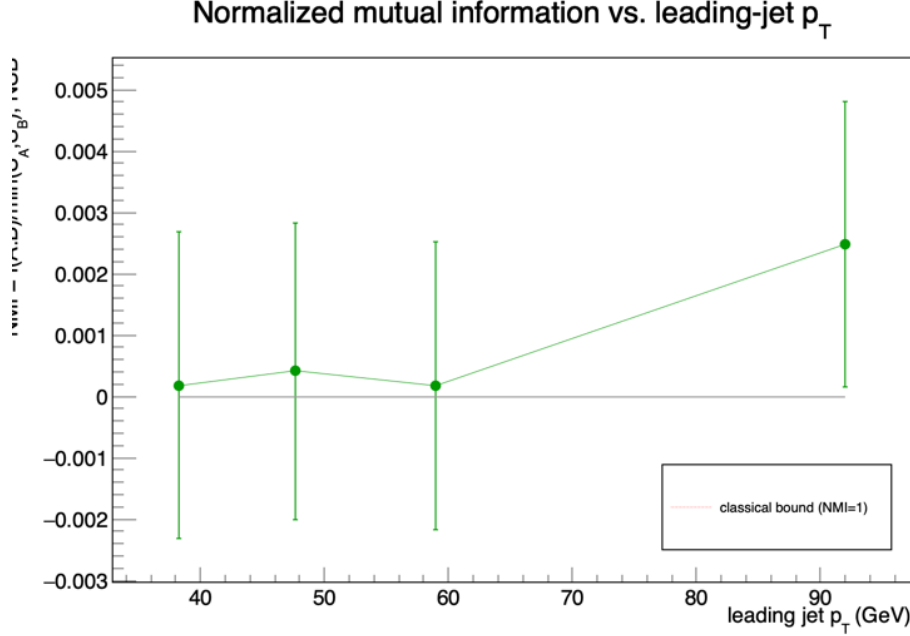


Figure 20: PYTHIA 8 (ROOT implementation): $NMI = I(A:B)/\min(S_A, S_B)$ (NSB) vs. leading-jet p_T . All values are far below the classical bound $NMI = 1$, as expected for a classical event generator.

individual p_T bin, particularly the highest- p_T , lowest-statistics bins where a genuine signal would be most interesting and least well-measured. A p_T -differential extension of the closure test has been implemented: for each p_T bin independently, fit a bin-specific Gamma-Poisson model to that bin’s own events (not the whole-sample fit), generate a bin-specific pseudo-truth, and run the closure procedure at that bin’s actual event count – directly answering “is NSB’s uncertainty trustworthy in *this* bin” rather than assuming the whole-sample answer applies uniformly.

Figure 26 shows the result, in the same four p_T bins used throughout Sec. 9.2 (p_T centers $\approx 38, 48, 59, 92$ GeV). The pattern is notably different from, and in one respect more encouraging than, the whole-sample result of Sec. 10.2:

- **Pull means** for $I(A:B)$ and NMI are small and negative throughout (≈ -0.12 to -0.17 in the three lower- p_T bins, shrinking to ≈ -0.04 in the highest- p_T bin) – markedly smaller in magnitude than the whole-sample ≈ -1 found in Sec. 10.2. $S(B|A)$ ’s pull mean is somewhat larger at low p_T (≈ -0.29) but trends toward zero as p_T increases, reaching ≈ 0 in the highest- p_T bin.
- **Pull standard deviations** for $I(A:B)$ and NMI rise from ≈ 0.58 (lowest- p_T bin, i.e. *over*-covering – the bootstrap σ is if anything too conservative there) through ≈ 0.80 and ≈ 0.95 to ≈ 0.92 in the highest- p_T bin – staying at or below 1 throughout, in contrast to the whole-sample 1.5–1.7 under-covering. $S(B|A)$ ’s pull standard deviation is close to 1 in every bin ($\approx 1.02, 1.00, 1.11, 0.92$) – the best-calibrated of the three quantities here, and encouragingly close to ideal given $S(B|A)$ is the quantity that would carry an actual discovery claim.

This bin-by-bin result being better-calibrated than the whole-sample one is itself a meaningful, self-consistent finding, not merely a nicer number: Sec. 10.4’s entire motivation was that the

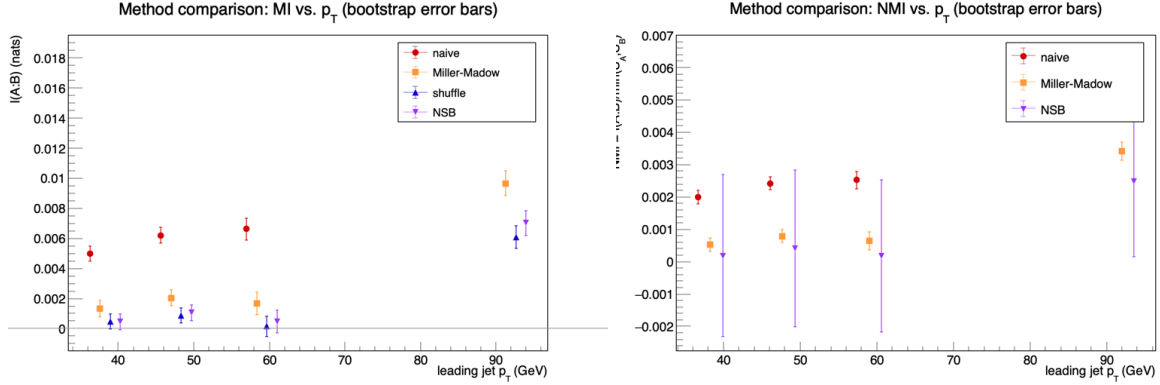


Figure 21: The bug, illustrated: $I(A:B)$ method comparison (left, unaffected) alongside the *pre-fix* NMI method comparison (right) from the same pipeline run. NSB’s NMI error bars are visibly far wider, relative to the central values, than I ’s – despite both being derived from the same underlying quantities on the same events. This discrepancy was the symptom that led to identifying the mixed-methodology bug described in the text.

whole-sample Gamma-Poisson fit has to compromise across a range of p_T -dependent correlation structures at once (Sec. 9.2), whereas each bin-specific fit only has to describe its own, comparatively homogeneous sub-population. A better-calibrated per-bin result is exactly what that design difference predicts, and is evidence the whole-sample closure test’s miscalibration (Sec. 10.2) is at least partly a model-fit-quality effect specific to forcing one 3-parameter model to span the full p_T range, rather than a property of NSB’s bootstrap uncertainty in general. That said, this is a single validation dataset at one set of statistics; whether this pattern holds at STAR’s actual 200 GeV kinematics and statistics (Sec. 12) is not yet established and should not be assumed.

Even with this more encouraging per-bin picture, no bin here reaches pull std = 1 exactly, and the lowest- p_T bin’s over-covering for I/NMI (≈ 0.58) is itself a $\sim 40\%$ deviation from ideal calibration, in the conservative direction. The practical recommendation of Sec. 10.2 therefore still stands as the safe default – treat raw bootstrap σ with caution and consult a closure test at the actual bin/statistics in use – with this section’s result showing that the correction, at least for this validation sample, tends to be smaller and bidirectional per bin rather than a uniform large under-coverage.

11 A distinct systematic: trivial kinematic MI, and its removal by mixed-event subtraction

Everything so far corrects *estimator* bias: the gap between a finite-sample MI estimate and the true MI of the sample’s generating distribution. This section addresses a different problem, upstream of estimation entirely – a sense in which the *true* MI being estimated is itself partly an artifact of the measurement’s finite p_T binning, not a real dynamical correlation.

11.1 The problem

Both jets’ multiplicities depend on their own p_T (Sec. 9.2), and $p_{T,1}$, $p_{T,2}$ are themselves correlated through momentum balance. This is a common-cause structure: even if N_A and N_B were completely independent *conditional on* the event kinematics, integrating over any p_T window of finite width

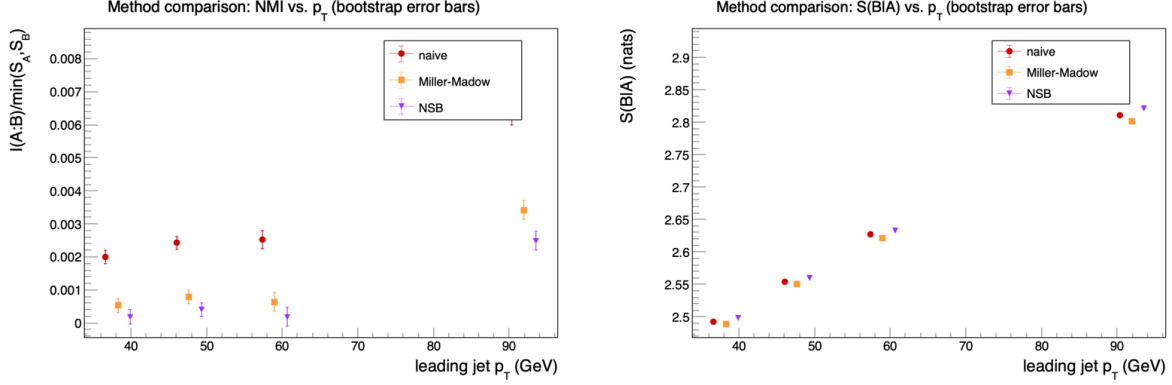


Figure 22: Post-fix confirmation, same pipeline and dataset as Fig. 21: NMI method comparison (left) now shows NSB error bars comparable in relative size to naive/Miller–Madow’s, in line with $I(A:B)$ ’s panel in Fig. 21 – the mixed-methodology discrepancy is gone. $S(B|A)$ method comparison (right, never affected by this particular bug) shown alongside for reference, confirming consistent bootstrap behavior across all three quantities.

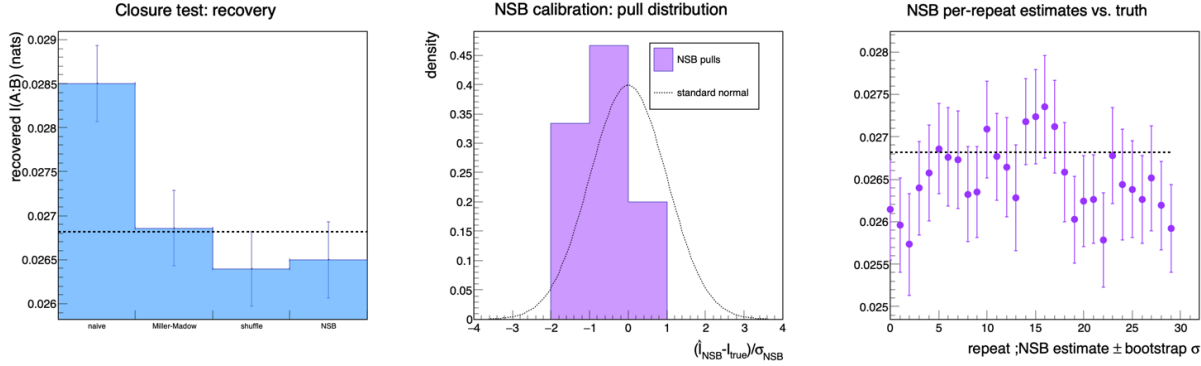


Figure 23: Whole-sample closure test, $I(A:B)$: recovery across methods (left), NSB pull distribution vs. standard normal (center), per-repeat NSB estimates vs. the fitted-model pseudo-truth (right).

induces real, nonzero $I(A:B)$ – not estimator bias, genuine correlation, but kinematically trivial and present in a purely classical generator with no dynamical jet–jet correlation at all. This does not vanish for any finite bin width, only shrinks as the bin narrows; the quantity one actually wants is closer to the conditional mutual information $I(N_A:N_B | p_T)$, which finite-width binning only approximates from above. Notably, this problem does *not* affect the quantum-witness test itself ($S(B|A) < 0$ or $\text{NMI} > 1$): trivial kinematic correlation can only push a measurement deeper into the classical region, never fake a violation. It matters specifically for interpreting the *magnitude* of a measured $I(A:B)$ as evidence of dynamical (as opposed to purely kinematic) correlation.

11.2 Mixed-event subtraction

For each event, jet A’s multiplicity and kinematics are kept as measured; jet B’s multiplicity is replaced with that of a *different* event’s jet B, selected by k -nearest-neighbor matching in $p_{T,2}$ (default $k = 10$). Mixed pairs then reproduce the same kinematic structure as the same-event sample, but with N_B drawn conditionally independent of N_A given (matched) kinematics – so the mixed sample’s own MI, I_{mix} , directly estimates the trivial kinematic contribution. The beyond-

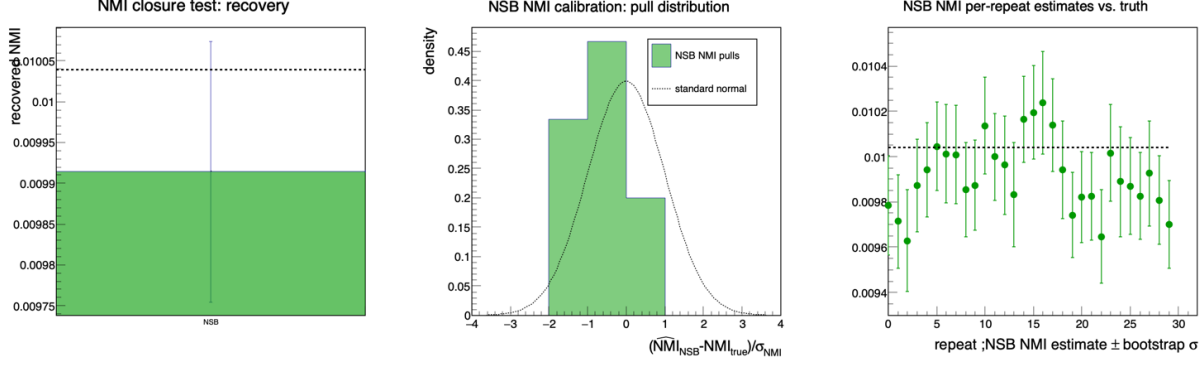


Figure 24: Whole-sample closure test, NMI: same three panels as Fig. 23, for the normalized witness.

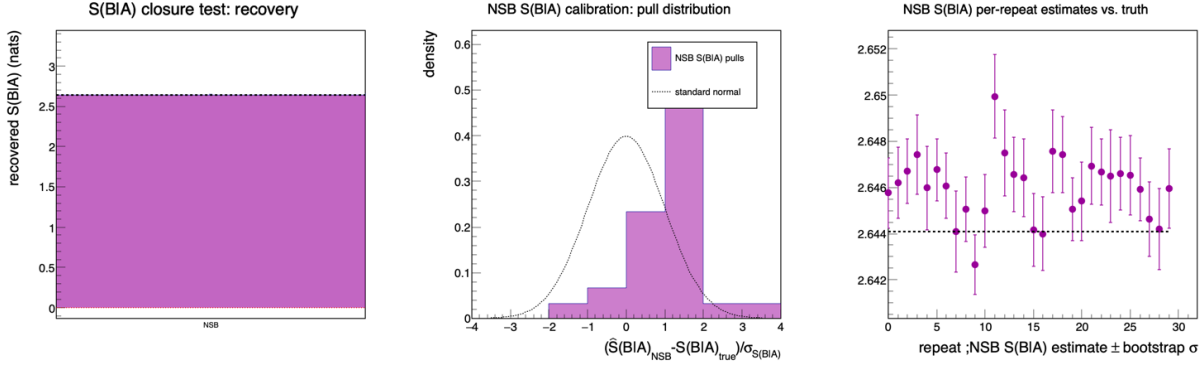


Figure 25: Whole-sample closure test, $S(B|A)$: same three panels as Fig. 23, for the conditional-entropy witness. Note the positive pull skew and tighter spread compared to I/NMI (Fig. 23–24).

kinematics signal is

$$I_{\text{signal}} = I_{\text{same}}^{\text{corrected}} - I_{\text{mix}}^{\text{corrected}}, \quad (17)$$

with bias correction applied to *each term separately* – not to the already-subtracted difference, which would double-correct, since the leading naive bias $\approx (K_{AB} - K_A - K_B + 1)/(2N)$ largely cancels between the two terms already (same marginals and same N by construction). Uncertainties are obtained by a *joint* bootstrap of the full difference: each resample rebuilds both the same-event pairs and the mixed pairs (re-matching within the resample) and recomputes $I_{\text{same}} - I_{\text{mix}}$ for all four methods, never propagating the two terms’ errors separately – the same lesson as Sec. 9.8, and more important here since the two terms share events and are strongly correlated. The matching resolution k is itself a systematic: any residual within-window p_T variation remains unsubtracted trivial MI, and varying k and checking stability of I_{signal} is the natural check.

11.3 Relation to shuffle-subtraction

The two corrections are not redundant, though they are the same operation at different resolutions. Shuffling permutes N_B labels *within the analysis sample*, which realizes the product of that sample’s own marginals – exactly what mixed-event pairing gives in the limit of no kinematic matching at all ($k \rightarrow$ the full sample). Formally: *shuffle-subtraction is mixed-event subtraction with matching resolution equal to the analysis bin*. What differs is the null each realizes, and so what each

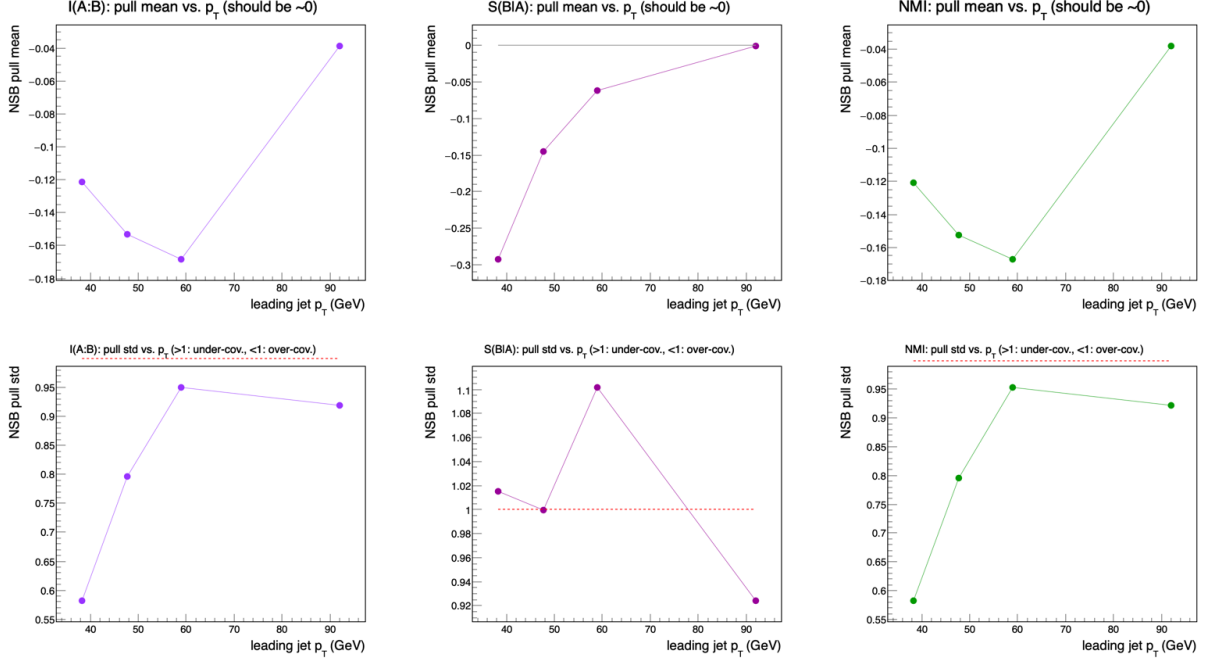


Figure 26: p_T -differential closure test: NSB pull mean (top row) and standard deviation (bottom row) vs. leading-jet p_T , for $I(A:B)$ (left), $S(B|A)$ (center), and NMI (right). Red dashed line marks perfect calibration (pull std = 1).

subtraction removes: shuffling’s null has true MI exactly zero, so it corrects only finite-sample estimator bias and deliberately leaves all real correlation – including the trivial kinematic part – intact. Fine-matched mixing’s null has nonzero true MI (the trivial kinematic part), so it removes a genuine physical contribution. The two are complementary, not alternatives: bias correction (via shuffle, Miller–Madow, or NSB) is applied to each of I_{same} and I_{mix} individually, and the kinematic subtraction is taken between the two corrected numbers.

11.4 Validation: bias near-cancellation and closure test

Figure 27 first validates the construction on a purpose-built synthetic dataset with $N_A \perp N_B \mid p_T$ exactly by construction (a shared momentum-balance kinematic correlation with no dynamical jet–jet correlation added) – a case where I_{signal} should recover zero. The raw same-event MI is entirely trivial (≈ 0.034 – 0.039 nats depending on method) yet the mixed-event difference is consistent with zero for every one of naive, Miller–Madow, shuffle, and NSB. The near-cancellation predicted above is directly visible: the *raw* naive same-event and mixed-event estimates individually carry substantial bias (0.0386 vs. a true value near zero), but naive-minus-naive (-0.00132) lands almost on top of NSB-minus-NSB (-0.00140) – the bias has canceled between the two terms to within the bootstrap uncertainty, before any explicit correction was applied at all.

This motivated a dedicated closure test for the subtraction itself, distinct from Sec. 10’s closure tests for the raw estimators. The model resamples $(p_{T,1}, p_{T,2})$ pairs directly from real data (reproducing the true kinematic correlation exactly, no spectrum model needed), fits $\lambda_j(p_T) = a_j + b_j \ln p_T$ to each jet’s real multiplicity– p_T profile, and injects a controllable *genuine* beyond-kinematics correlation via a shared latent $g \sim \text{Gamma}(1/s^2, s^2)$ multiplying both jets’ rates ($s = 0$: exactly conditionally independent, true signal exactly zero; $s > 0$: a known injected signal). Pseudo-truth

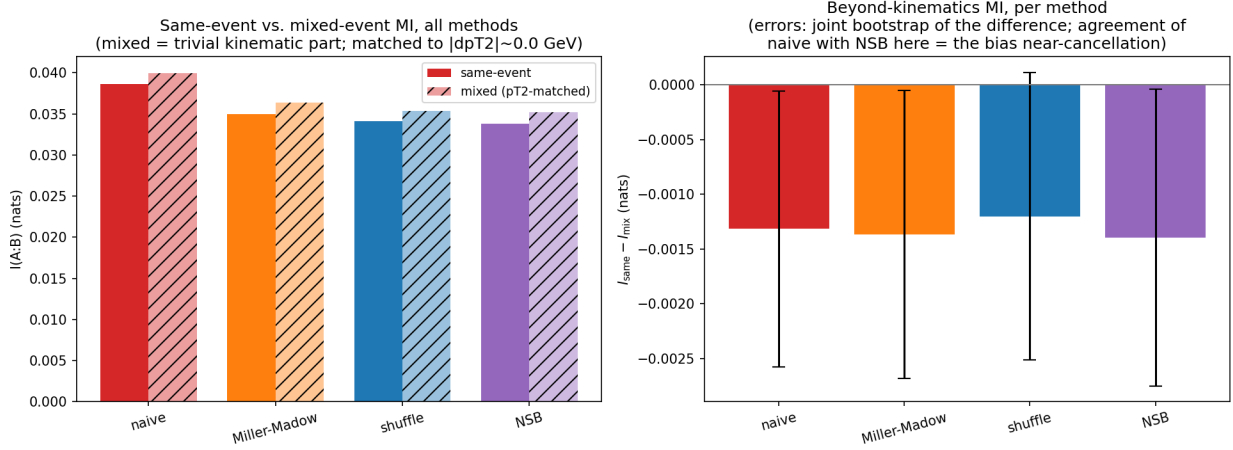


Figure 27: Mixed-event subtraction QA on a synthetic dataset with $N_A \perp N_B \mid p_T$ by construction. Left: same-event vs. $p_{T,2}$ -matched mixed-event $I(A:B)$, all four methods – both are dominated by the trivial kinematic contribution. Right: the per-method difference with jointly-bootstrapped errors, consistent with zero throughout; naive-minus-naive agreeing with NSB-minus-NSB despite the raw naive terms’ visible individual bias is the near-cancellation described in the text.

is obtained from a large reference sample analyzed with the identical matched-mixing procedure, so the truth is defined for exactly the estimand the method targets.

Unlike the reduced-statistics smoke test reported in an earlier revision of this note, Figs. 28–29 show the closure test run at full statistics (30 repeats, 30-fold joint bootstrap per repeat) directly on the real 14 TeV validation sample’s own fitted kinematics and multiplicity profile – both scenarios required by the design (Sec. 11): the *null* ($s = 0$), the essential false-positive control before claiming any beyond-kinematics signal on real data, and an injected-signal case ($s = 0.15$).

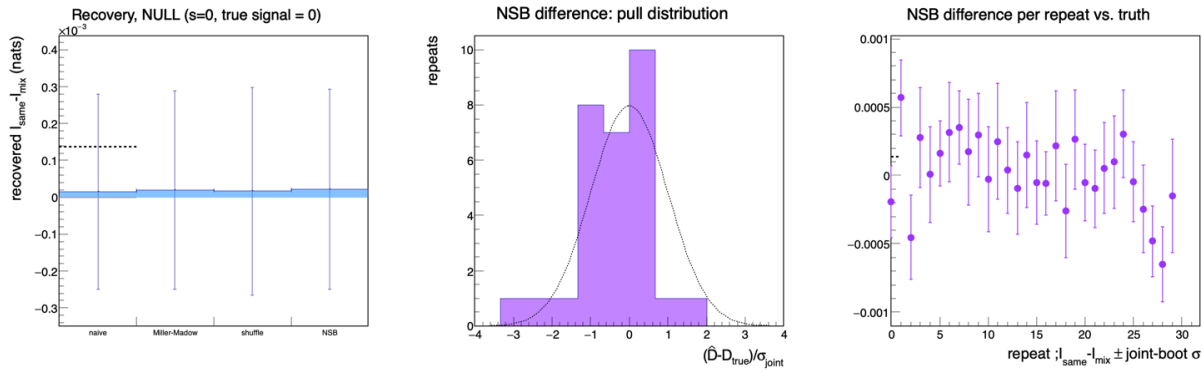


Figure 28: Closure test of the mixed-event subtraction on the real 14 TeV validation sample’s fitted kinematics, null scenario ($s = 0$), full statistics (30 repeats). Left: recovered $I_{\text{same}} - I_{\text{mix}}$ per method vs. the (near-zero) pseudo-truth. Center: NSB-difference pull distribution against standard normal. Right: per-repeat NSB estimates with joint-bootstrap error bars vs. the true value.

In the null scenario the pseudo-truth itself is small but not exactly zero ($\approx 1.4 \times 10^{-4}$ nats) – expected, since it is estimated by the naive estimator on a large but finite reference sample and carries that estimator’s own residual $\mathcal{O}(1/N)$ noise, not a flaw in the null construction. All four

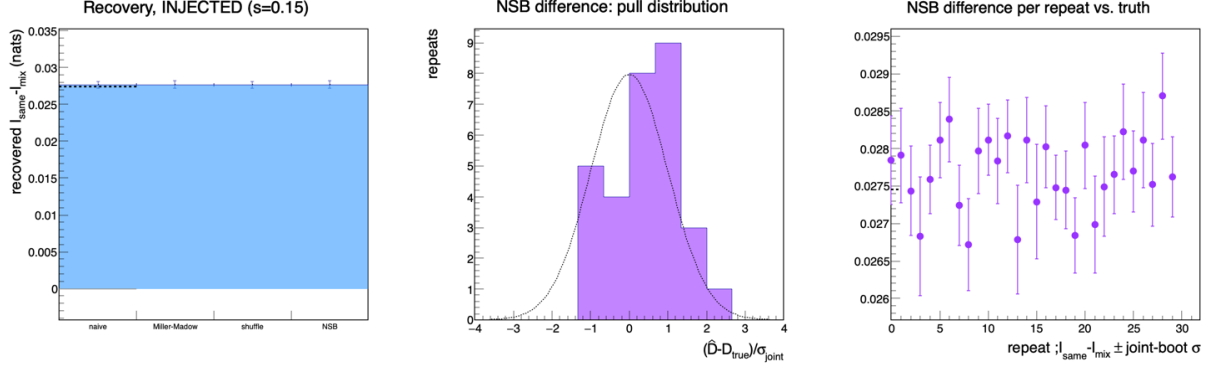


Figure 29: Same as Fig. 28, injected-signal scenario ($s = 0.15$, true difference ≈ 0.0275 nats).

methods recover values consistent with this truth within their (comparatively large, since the true difference is itself tiny) joint-bootstrap uncertainties, and the NSB-difference pull distribution does not show the pronounced skew seen in the raw-estimator whole-sample closure tests (Sec. 10.2). In the injected-signal scenario, all four methods recover the true ≈ 0.0275 nats difference closely and consistently, again without the earlier tests' marked under-coverage. Taken together, the mixed-event *difference* appears better-calibrated at these statistics than the raw single-term estimators were – plausibly because the joint bootstrap directly captures the strong correlation between the same-event and mixed-event terms (they share events), which the raw single-quantity closure tests had no analogue of. This is an encouraging empirical finding rather than a proven general property, and should not be assumed to hold automatically at STAR's real kinematics and statistics without rerunning this same check there.

11.5 Application to the real validation sample

With the method validated, it was applied to the actual 14 TeV PYTHIA 8 validation sample used throughout Sec. 9 – answering the question Sec. 11's design was motivated by: how much of the $I(A:B)$ measured elsewhere in this note is trivial kinematic correlation, versus a beyond-kinematics component.

Figure 30 shows the whole-sample result. Same-event $I(A:B) \approx 0.0084\text{--}0.0112$ nats depending on method (naive highest, as expected from its known upward bias), dropping to $\approx 0.0055\text{--}0.0082$ nats for the $p_{T,2}$ -matched mixed sample. The difference is ≈ 0.0030 nats, and – notably – essentially *the same* value across all four methods, with joint-bootstrap uncertainty $\approx \pm 0.0004\text{--}0.0005$ nats. Roughly 60–70% of the raw same-event MI is removed as trivial kinematic correlation (depending slightly on method); the remaining ≈ 0.003 nats is a genuine, nonzero beyond-kinematics signal, not an artifact of any one estimator's bias.

Since PYTHIA 8 has no mechanism for quantum entanglement between separately hard-scattered jets, this remaining correlation must be classical in origin. The most likely candidate, anticipated already in Sec. 11's design discussion, is shared underlying-event activity – multiple-parton interactions, beam-remnant particles, and color reconnection all act coherently on *both* jets within the same event, and matching mixed partners only on jet-B's p_T does nothing to remove event-level (as opposed to jet-level) shared structure. This is consistent with, not a contradiction of, the subtraction's design: the method isolates correlation beyond simple p_T -driven kinematics, and underlying-event effects are exactly the kind of genuine (if unglamorous) dynamical correlation that description was always going to include.

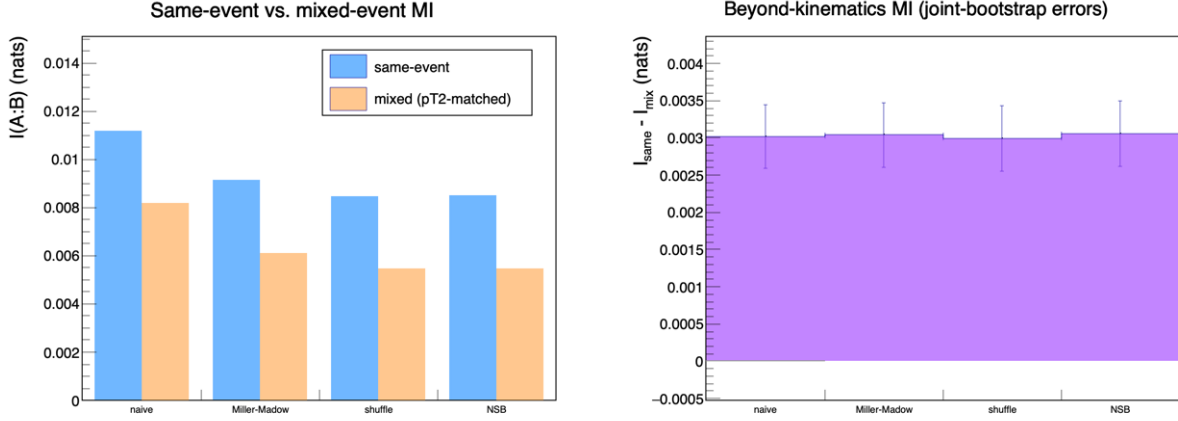


Figure 30: Mixed-event subtraction applied to the real 14 TeV validation sample (whole sample). Left: same-event vs. $p_{T,2}$ -matched mixed-event $I(A:B)$, all four methods. Right: the per-method difference with jointly-bootstrapped errors – a nonzero beyond-kinematics component, consistently recovered across all four estimators.

The p_T -differential breakdown (Fig. 31) is less clean, as expected given each bin carries roughly a quarter of the whole sample’s statistics: the two lower- p_T bins are consistent with zero, the third bin shows a small positive value (≈ 0.0009 – 0.0011 nats), and the highest- p_T bin shows the largest value (≈ 0.0015 – 0.0022 nats) but also by far the largest uncertainty. All four methods agree closely within each bin, which is reassuring for the subtraction’s internal consistency, but no individual bin here should be read as an independently significant detection – the whole-sample result in Fig. 30, with its higher statistics and tighter relative uncertainty, is the more trustworthy number at this stage. A closure test run specifically at each bin’s statistics (the p_T -differential analogue of Sec. 10.4, not yet built for the mixed-event method) would be needed before quoting a per-bin significance with confidence.

11.6 Status and outlook

The subtraction machinery, its bias-cancellation property, a closure-test framework validating it at full statistics on the real sample’s own kinematics, and an application to the real validation sample itself are now all complete (Figs. 27–31). What remains: a p_T -differential closure test for the mixed-event difference specifically (paralleling Sec. 10.4’s extension of the raw-estimator closure test), needed before the per-bin breakdown in Fig. 31 can support a significance claim in any individual bin; and, longer-term, isolating the underlying-event contribution more directly (e.g. by matching on additional event activity variables, not just jet-B’s p_T) if a cleaner separation of kinematic, underlying-event, and (eventually, on real detector data) any genuinely anomalous component is wanted.

12 Overall conclusions and next steps

The toy Monte Carlo study (Secs. 4–7) establishes the estimator methodology and its failure modes under controlled, known conditions. Section 9 shows the same pipeline, unmodified, correctly recovers physically-expected behavior (NBD-like overdispersion, MLLA-consistent multiplicity growth

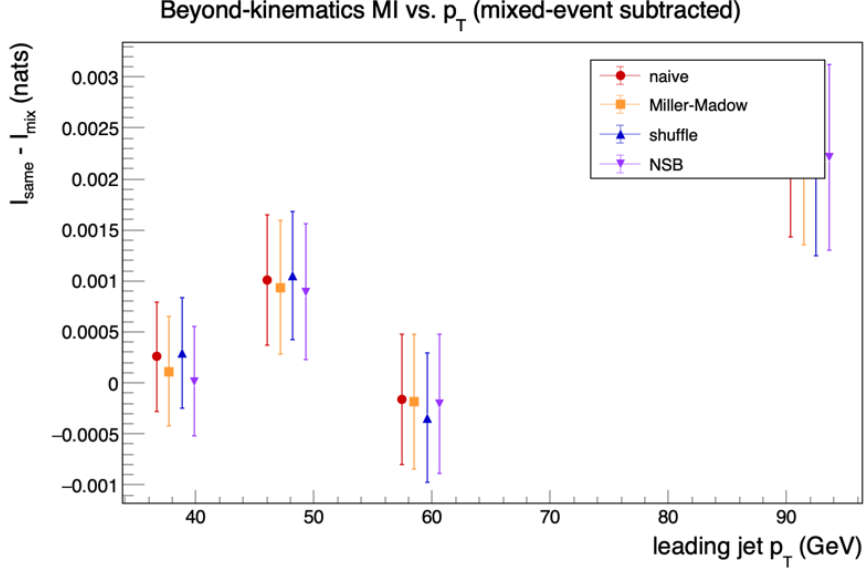


Figure 31: Beyond-kinematics MI vs. leading-jet p_T (mixed-event subtracted), real validation sample, all four methods.

with p_T , the same finite-sample bias shape) on realistic QCD Monte-Carlo output where no such structure was hand-built in – a meaningful, if indirect, validation that the pipeline is measuring real structure rather than an artifact of the toy model’s specific construction. An independent ROOT/C++ re-implementation (Sec. 9.7) cross-validates the Python results and adds NMI, the scale-comparable normalized version of the witness (Sec. 2) needed to compare measurements across p_T bins, run periods, or collision energies on a common, bounded scale.

That cross-validation also surfaced a genuine implementation bug (Sec. 9.8): a bootstrap-based derived quantity (NMI) picked up its uncertainty from an inconsistent, more conservative analytic formula rather than the same resampling procedure used for $I(A:B)$ and $S(B|A)$ in the same function. The fix, and more importantly the general principle it illustrates – every derived quantity built from one bootstrap procedure must be resampled together, not mixed with a separately-derived analytic error – is recorded in full since it is a pitfall relevant to any bootstrap-based entropy/MI pipeline, not just this one.

Section 10 closes the gap the previous revision of this note flagged as open: a closure test calibrated to resemble the real PYTHIA 8 sample (Gamma-Poisson model fit to the real data’s marginals and correlation strength, huge reference sample for a pseudo-truth, repeated closure at the real dataset’s actual statistics) finds that NSB’s bootstrap uncertainty, at this sample’s (N , alphabet), is **under-covering by a factor of roughly 1.5–1.7**, with a mean bias of order -1σ and a pull distribution that is skewed rather than symmetrically widened. This is a quantitative, load-bearing result for any future significance claim: a naive 3σ excess computed from raw bootstrap σ at comparable statistics should be read as $\sim 1.8\text{--}2\sigma$ once this calibration factor is accounted for. Per Sec. 10, no correction has been baked into the pipeline itself – this factor is reported here as a measured property of the estimator at these statistics, to be applied explicitly and cited whenever a significance is quoted, rather than silently folded into the numbers.

The p_T -differential extension of the closure test (Sec. 10.4) is now complete for this validation sample, and adds an important nuance: bin-by-bin, NSB’s pull standard deviations sit much closer to the ideal value of 1 ($\approx 0.58\text{--}0.95$ for $I(A:B)$ and NMI; $\approx 0.92\text{--}1.11$ for $S(B|A)$) than the

whole-sample result above, consistent with each bin’s own Gamma-Poisson fit describing a more homogeneous sub-population than the one compromise fit spanning the whole p_T range has to. This suggests the whole-sample under-coverage found in Sec. 10.2 is at least partly a consequence of fitting one model across a range of p_T -dependent structure, rather than an intrinsic property of NSB’s bootstrap procedure – but this is one validation dataset at one set of statistics, and the pattern should be re-checked, not assumed, at STAR’s actual kinematics. The dedicated whole-sample pull-distribution check for $S(B|A)$ itself (Sec. 10.3 – distinct from its per-bin counterpart just described, and the single most directly load-bearing calibration number for a discovery claim) is now complete: its pull distribution is positively skewed (a modest over-estimate, opposite in sign from I/NMI ’s under-estimate) but visibly tighter, closer to nominal calibration than I/NMI ’s 1.5–1.7 factor. That the two quantities calibrate differently on the same sample is itself informative – a single blanket correction factor would not be appropriate across all three observables.

Section 11 addresses a separate concern from estimator calibration: any finite p_T window induces real but kinematically trivial MI through the momentum-balance correlation between the two jets’ p_T ’s. A mixed-event subtraction removes this by construction and exhibits a useful property found rather than assumed: the leading finite-sample bias very nearly cancels between the same-event and mixed-event terms, so the subtracted difference is close to unbiased even before explicit correction – confirmed both on a controlled synthetic dataset and by a full-statistics closure test run directly on the real sample’s fitted kinematics (Sec. 11), where the mixed-event difference showed better calibration than the raw single-term estimators of Sec. 10. Applied to the real 14 TeV validation sample, roughly 60–70% of the raw whole-sample $I(A:B)$ is trivial kinematic correlation; the remaining ≈ 0.003 nats, consistent across all four estimator methods, is a genuine beyond-kinematics signal most plausibly attributable to shared underlying-event activity (Sec. 11.5) rather than anything exotic – exactly the kind of confound this correction exists to separate out before any dynamical or quantum interpretation of a measured $I(A:B)$ magnitude is warranted.

Remaining steps toward the actual STAR measurement: (i) repeat Sec. 9 at $\sqrt{s} = 200$ GeV with STAR-appropriate jet kinematics rather than the wide-dynamic-range 14 TeV validation kinematics used here; (ii) apply the detector-level acceptance and tracking-efficiency model (already implemented in the generator, not yet exercised in this note) and repeat the full analysis chain; (iii) rerun the whole-sample and p_T -differential closure tests (Secs. 10.2–10.4) at the actual statistics of (i)–(ii), obtaining calibration factors specific to the real analysis rather than relying on the 14 TeV validation sample’s factors as a proxy; (iv) rerun the mixed-event subtraction and its closure test (Sec. 11) at STAR kinematics and statistics – done so far only on the 14 TeV validation sample – including the still-missing p_T -differential closure test for the subtracted quantity, needed before any per-bin beyond-kinematics result can carry a trustworthy significance; (v) once (i)–(iv) are complete, apply the identical pipeline to real STAR data, with the sensitivity/detection-threshold machinery of Sec. 7 and all applicable closure-test calibration factors applied explicitly, before unblinding, to state precisely how large a genuine quantum-correlation excess would need to be to constitute a significant detection at the actual achieved statistics.